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PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 17

INTRODUCTION TO MOTION OF CONNECTED PARTICLES

SUGGESTED TIME: 9 PERIODS

IN THIS CHAPTER

This chapter introduces the dynamics of two or more bodies linked by strings, rods or contact, moving together under gravity, friction and applied forces. Students learn to model connected systems (smooth and rough surfaces, inclined planes, pulleys and Atwood's machine), to compute accelerations and tensions, and to solve real-life problems using both Newton's laws and energy methods.

- By the end of this chapter I can model a system of connected particles with clear assumptions (light inextensible string, smooth pulley, particle model).
- By the end of this chapter I can draw separate force diagrams for each particle of a connected system.
- By the end of this chapter I can apply Newton's second law to each particle and solve the simultaneous equations for acceleration and tension.
- By the end of this chapter I can analyse motion of connected particles on smooth horizontal surfaces.
- By the end of this chapter I can include limiting and non-limiting friction ($F = \mu R$) in connected-particle problems on rough surfaces.
- By the end of this chapter I can resolve forces on an inclined plane and analyse particles connected over a pulley on slopes.

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METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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2. Section 2: Motion on Smooth and Rough Horizontal Surfaces
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4. Section 4: Atwood's Machine and Multi-Stage Motion
5. Section 5: Pulley Systems with More Than Two Masses and Variable Acceleration
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CHAPTER OBJECTIVES

- By the end of this chapter I can model a system of connected particles with clear assumptions (light inextensible string, smooth pulley, particle model).
- By the end of this chapter I can draw separate force diagrams for each particle of a connected system.
- By the end of this chapter I can apply Newton's second law to each particle and solve the simultaneous equations for acceleration and tension.
- By the end of this chapter I can analyse motion of connected particles on smooth horizontal surfaces.
- By the end of this chapter I can include limiting and non-limiting friction ($F = \mu R$) in connected-particle problems on rough surfaces.
- By the end of this chapter I can resolve forces on an inclined plane and analyse particles connected over a pulley on slopes.
- By the end of this chapter I can solve Atwood's machine problems, including the force on the pulley support.
- By the end of this chapter I can handle systems with more than two masses and multi-stage motion (string going slack, particle hitting the ground).
- By the end of this chapter I can use the work–energy principle and conservation of mechanical energy as an alternative method for connected systems.

- By the end of this chapter I can interpret and solve real-life connected-particle problems in the style of GCE Paper 2 questions.

PREREQUISITES

- Newton's three laws of motion and F = ma (Lower Sixth mechanics)
- Resolving forces into components and equilibrium of a particle
- Friction: coefficient of friction, $F \leq \mu R$, limiting equilibrium
- Kinematics with constant acceleration: the suvat equations
- Work, kinetic energy, potential energy and the work–energy principle

1 SECTION 1: MODELLING CONNECTED PARTICLES AND TENSION IN STRINGS

At the Mile 4 motor park in Limbe, a mechanic uses a rope passing over a beam to lift a heavy engine out of a lorry: the engine goes down slowly while a counterweight of scrap metal rises. Why does the engine not crash to the ground, and how heavy must the counterweight be? On the Bamenda escarpment, a pickup tows a broken-down taxi up a steep slope with a chain — will the chain hold? These everyday situations all involve **connected particles**, and by the end of this chapter you will be able to predict their motion, the forces in the connectors, and the energy involved, exactly as required at the GCE Advanced Level.

In this first section we build the language of the whole chapter: what a connected-particle system is, the modelling assumptions that make it tractable, the nature of **tension**, and a five-step routine that turns any such problem into a pair of simultaneous equations.

1.1 What is a Connected-Particle System?

So far in mechanics you have mostly studied a single particle moving under given forces. In real life, bodies rarely move alone: a locomotive pulls wagons, a crane cable lifts a container, two climbers are tied together by a rope. Whenever two or more bodies are linked by a string, rope, chain, rod or tow-bar, the motion of one body constrains the motion of the others: they cannot move independently.

DEFINITION **Connected particles**

Two or more bodies are called **connected particles** when they are linked by a connector (string, rope, chain, rod or tow-bar) in such a way that the motion of one body determines, wholly or partly, the motion of the other(s). In the standard model, all particles attached to the same taut, inextensible string move with the same speed and the same magnitude of acceleration.

Think of the mechanic's rope in Limbe: if the engine descends by 30 cm, the counterweight must rise by exactly 30 cm, because the rope does not stretch. The two motions are locked together — that is the whole point of the model.

1.2 Modelling Assumptions

Real ropes have mass, stretch a little, and rub on rough pulleys. To obtain clean mathematics we idealise. Each assumption below has a precise physical consequence that you must be able to state in an examination.

KEY POINTS **The standard modelling assumptions**

- Particle model: each body is treated as a point mass. Its size, shape and rotation are

ignored; all forces act at a single point.

- Light string (massless): the string has no mass, so no net force is needed to accelerate the string itself. Consequence: the **tension** has the same value at every point of the string. (Proof: for a massless string, $F_{\text{net}} = ma = 0$, so the forces at its ends are equal and opposite.)
- Inextensible string: the string does not stretch. Consequence: the distances moved, the speeds and the magnitudes of the accelerations of all particles attached to the same taut string are equal.
- Smooth pulley or peg: there is no friction in the pulley (and the pulley is light). Consequence: the tension is the same on both sides of the pulley; the pulley only changes the direction of the force.

WARNING **Inextensibility is a kinematic condition**

A common error is to treat the two particles as independent and give them different accelerations. If the string is taut and inextensible, this is impossible: the string would have to stretch or break. Always write one single unknown a for the common magnitude of acceleration, and one single unknown v for the common speed.

1.3 Tension in Strings

Take a taut string and pull it at both ends. Every small piece of the string is pulled by its neighbours; this internal pulling force, transmitted along the string, is the **tension**.

DEFINITION **Tension**

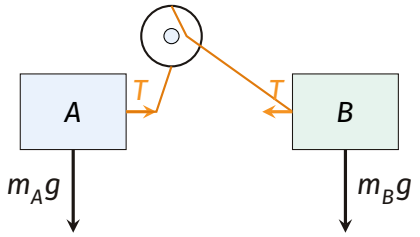
The **tension** T in a taut string is the pulling force that the string exerts on each body attached to it, directed along the string, away from the body (a string can only pull). If the string is light and passes over a smooth pulley, then T has the same value throughout the string.

Why does a string always pull and never push? Because a flexible string cannot sustain compression: push it and it simply buckles and goes slack. When a string goes slack, $T = 0$ and the particles cease to be connected — this is a classic examination trap.

PROPERTY **Newton's third law and tension**

The forces exerted by a taut string on the two bodies at its ends form a Newton third-law

pair: they are equal in magnitude and opposite in direction. If the string pulls body A with force T towards the string, it pulls body B with force T towards the string at the other end. (Proof not examinable; it follows directly from the light-string assumption and $F = ma$ applied to the massless string.)



Free-body diagrams of two particles connected by a light inextensible string over a smooth pulley. The tension T has the same magnitude on both sides.

Strings versus rods and tow-bars.

A string, rope or chain can only pull: the force in it is always a tension ($T \geq 0$), and it may go slack. A rigid **rod** or **tow-bar** can also push: the force in it can be a tension (pulling) or a **thrust** (pushing, sometimes written $T < 0$ or stated separately as a thrust). When a car tows a trailer with a rigid tow-bar and then brakes hard, the bar may be in thrust, pushing the trailer back. Always read the question to see which connector is used.

1.4 The Five-Step Routine

Every connected-particle problem in this chapter is solved by the same disciplined routine.

METHOD **Solving a connected-particle problem**

1. State the assumptions: particles, light inextensible string (or light rigid rod), smooth pulley (or smooth/rough surface, as given).
2. Draw a separate free-body diagram for each particle, showing only the forces acting on that particle: weight, normal reaction, tension(s), applied forces, friction.
3. Choose the direction of motion for each particle (consistent with the string) and mark the common acceleration a.
4. Write Newton's second law $F = ma$ for each particle, resolved along its own direction of motion. (Alternatively, write the equation for the whole system to find a directly, then use one particle's equation to find T.)
5. Solve the simultaneous equations for a and T (and check the sign of T: for a string, $T \geq$

0; if $T < 0$, the string is slack and the model breaks down).

WARNING **One diagram per particle!**

The most frequent mistake at GCE is to draw both particles on a single diagram and to mix internal forces (the tensions) with external forces (weights, driving forces, resistances). For the whole system the internal tensions cancel out, but for each particle separately the tension is essential. Always draw two distinct diagrams and write two distinct equations.

1.5 First Example: Two Particles Over a Smooth Fixed Pulley

Consider two particles of masses m_1 and m_2 ($m_1 > m_2$) tied to the ends of a light inextensible string passing over a smooth fixed pulley — the mechanic's situation in Limbe, idealised. The heavier mass m_1 descends, the lighter mass m_2 rises, both with acceleration of magnitude a. Applying the routine:

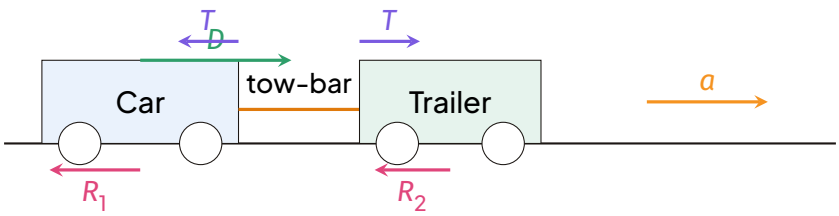
- Particle 1 (moving down, take down as positive): $m_1 g - T = m_1 a$.
- Particle 2 (moving up, take up as positive): $T - m_2 g = m_2 a$.

Adding eliminates T:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}.$$

Note that $a < g$: this is why the engine does not crash to the ground — the counterweight, through the tension, holds it back. The full quantitative study of this Atwood machine comes in Section 4.

1.6 Worked Example: A Car Towing a Trailer



Car towing a trailer: driving force D and resistance R_1 on the car, resistance R_2 on the trailer, and the tow-bar force T pulling the trailer forward and the car backward.

WORKED EXAMPLE 1 **Car towing a trailer on a horizontal road**

A car of mass 1200 kg tows a trailer of mass 400 kg along a straight horizontal road by means of a rigid tow-bar. The car's engine produces a driving force of 2400 N. Resistance forces of 600 N act on the car and 200 N on the trailer. Find (a) the acceleration of the system, (b) the force in the tow-bar, stating whether it is a tension or a thrust.

SOLUTION

Solution. Step 1 – Assumptions: car and trailer are particles; the tow-bar is light and rigid; the road is horizontal.
Step 2 – Diagrams: separate diagrams for car and trailer (see figure above).
Step 3 – Direction of motion: forwards, common acceleration a.
Step 4 – Newton's second law.
For the trailer (400 kg), the only horizontal forces are the tow-bar pull T and the resistance:

$$T - 200 = 400a. \tag{1}$$

For the car (1200 kg), the driving force $D = 2400$ N acts forwards, the resistance 600 N and the tow-bar reaction T act backwards:

$$2400 - 600 - T = 1200a. \tag{2}$$

Step 5 – Solve. Adding (1) and (2) gives the system equation:

$$2400 - 600 - 200 = 1600a \implies a = \frac{1600}{1600} = 1\,\text{m s}^{-2}.$$

From (1): $T = 200 + 400(1) = 600$ N.
Answers: (a) $a = 1\,\text{m s}^{-2}$; (b) the tow-bar is in tension of 600 N (it pulls the trailer forward and drags the car backward). Since $T > 0$, the bar is pulling, not pushing.

EXERCISE 1 **A pickup and a taxi**

A pickup of mass 1500 kg tows a broken-down taxi of mass 900 kg along a horizontal road with a light chain. The driving force is 3600 N; resistances are 750 N on the pickup and 450 N on the taxi. Find the acceleration and the tension in the chain. What happens to the tension if the pickup brakes gently?

SOLUTION **Exercise 1**

Newton's second law for the taxi: $T - 450 = 900a$. For the pickup: $3600 - 750 - T = 1500a$. Adding: $2400 = 2400a$, so $a = 1\,\text{m s}^{-2}$ and $T = 450 + 900 = 1350$ N. If the pickup brakes, the taxi pushes on the pickup; a chain cannot push, so it goes slack ($T = 0$) and the taxi

runs free until the chain tightens again.

DID YOU KNOW? **Why tow-bars instead of ropes?**

Road-safety rules in many countries require a rigid tow-bar rather than a rope for towing on fast roads: with a rope, the moment the towing vehicle slows down, the towed vehicle keeps going, the rope goes slack, and a collision follows. A rigid bar transmits thrust as well as tension, so the towing vehicle can help brake the towed one — exactly the string-versus-rod distinction of this section.

KEY POINTS **Key points of Section 1**

- Connected particles move with the same speed and same magnitude of acceleration when linked by a taut inextensible string.
- A light string over a smooth pulley has the same tension throughout; the pulley changes only the direction of the force.
- A string pulls only and may go slack ($T = 0$); a rod or tow-bar can also push (thrust).
- Always draw one free-body diagram per particle and write $F = ma$ for each, then solve simultaneously.
- Tension forces on the two bodies form a Newton third-law pair: equal magnitude, opposite direction.
- Check that $T \geq 0$ for a string; if $T < 0$ the string is slack and the particles move independently.

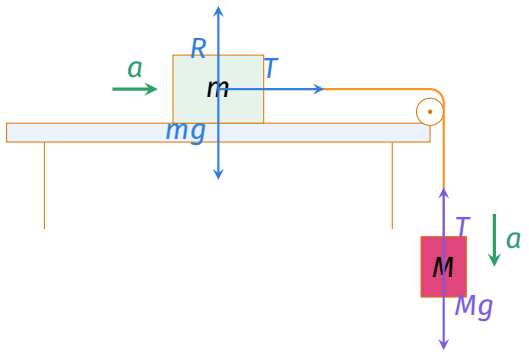
2 SECTION 2: MOTION ON SMOOTH AND ROUGH HORIZONTAL SURFACES

In Section 1 we learnt to model a system of connected particles: light inextensible strings, smooth pulleys, and the key fact that the **tension** is the same throughout a single string while both particles share the same **acceleration** magnitude. We now put these ideas to work in the most classic configuration of the GCE Advanced Level examination: one particle resting on a **horizontal table**, joined by a string passing over a pulley at the edge to a second particle hanging freely. We first treat the smooth table, then introduce friction, and finish with towing problems and the use of the constant-acceleration (suvat) formulae.

2.1 The Smooth Table with a Hanging Particle

EXPERIMENT **Setting up the situation**

Place a block of mass m on a polished table. Tie a light string to it, pass the string over a small pulley fixed at the edge of the table, and attach a mass M hanging freely. Release the system. The hanging mass falls, the block slides towards the edge, and both move together with the same acceleration because the string is inextensible. Our task is to predict that acceleration and the tension in the string.



Block of mass m on a smooth table, connected over a smooth pulley at the edge to a hanging mass M. Forces on each particle are shown.

Analysing each particle separately.

For the block on the table there is no vertical motion, so the vertical forces balance: $R = mg$. Horizontally the only force is the tension, so Newton's second law gives

$$T = ma. \tag{1}$$

For the hanging mass, taking the downward direction (the direction of its motion) as positive:

$$Mg - T = Ma. \tag{2}$$

Adding (1) and (2) eliminates T:

$$Mg = (M + m)a \quad \Longrightarrow \quad a = \frac{Mg}{M + m}, \qquad T = ma = \frac{Mmg}{M + m}.$$

METHOD **Smooth table: direct formulae**

For a hanging mass M driving a mass m on a smooth horizontal table:

1. Acceleration: $a = \dfrac{Mg}{M + m}$ — driving weight divided by total mass.

2. Tension: $T = \dfrac{Mmg}{M+m}$, obtained by substituting a into $T = ma$.
3. Always check: $T < Mg$ (the hanging mass is accelerating downwards, so its weight must exceed the tension).

This whole-system shortcut is valid whenever the string stays taut, the pulley is smooth, and no other horizontal force (such as friction) acts on the table particle.

WORKED EXAMPLE 2 **Smooth table: finding a and T**

A particle P of mass 2 kg rests on a smooth horizontal table. It is connected by a light inextensible string passing over a smooth pulley at the edge of the table to a particle Q of mass 3 kg hanging freely. The system is released from rest. Find the acceleration of the system and the tension in the string. Take $g = 9.8\ \text{m s}^{-2}$.

SOLUTION

Solution. For P (horizontal): $T = 2a$. For Q (downwards positive): $3g - T = 3a$. Adding:

$$3g = 5a \implies a = \frac{3 \times 9.8}{5} = 5.88\ \text{m s}^{-2}.$$

Then $T = 2a = 11.76$ N. Check: $T = 11.76 < 3g = 29.4$ ✓, and $3g - T = 17.64 = 3 \times 5.88$ ✓.

WARNING **The shortcut gives only a**

A very common error is to write $a = \dfrac{Mg}{M+m}$ and then claim $T = Mg$ or $T = Ma$. The whole-system equation contains no tension at all — the internal forces cancel. You must return to the equation of one particle ($T = ma$ or $Mg - T = Ma$) to find T. Notice also that $T = \dfrac{Mmg}{M+m}$ is less than both Mg and (usually) what candidates guess.

2.2 Introducing Friction: the Rough Table

A real table — a wooden desk in a Yaoundé classroom, for instance — is not smooth. When the block slides (or tends to slide), the surface exerts a frictional force opposing the motion.

PROPERTY **Normal reaction and the friction model**

For a particle on a horizontal surface with no vertical acceleration, resolving vertically gives

$$R = mg.$$

The frictional force satisfies $F \leq \mu R$, where μ is the coefficient of friction. When the body is moving (or in limiting equilibrium, on the point of moving),

$$F = \mu R,$$

directed opposite to the motion (or to the tendency of motion). If the body is at rest and not on the point of moving, F takes whatever value below μR is needed to maintain equilibrium.

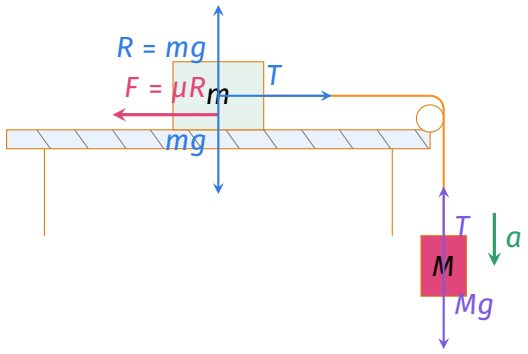
METHOD **Test first: will the system move at all?**

Before writing any equation of motion on a rough table:

1. Compute the maximum available friction $F_{\max} = \mu R = \mu mg$.
2. Compute the driving force, here the hanging weight Mg.
3. If $Mg \leq \mu mg$, the system does not move: friction adjusts to $F = Mg$ and the string tension is $T = Mg$ with $a = 0$.
4. If $Mg > \mu mg$, motion occurs; write the equations with $F = \mu R$ opposing the motion.

WARNING **Do not write $F = \mu R$ blindly**

Two mistakes are frequent. First, writing $F = \mu R$ for a smooth surface (where $F = 0$). Second, writing $F = \mu R$ when the body is at rest and the driving force is small: then $F < \mu R$, and assuming $F = \mu R$ can produce a negative acceleration — a contradiction telling you that your assumption of motion was wrong. The direction of F also matters: it always opposes the actual or impending motion, so it points away from the pulley here.



Rough table: friction $F = \mu R$ opposes the motion of the block towards the pulley, while vertically $R = mg$.

Equations of motion on the rough table.

Assuming motion occurs (block moving towards the pulley):

$$\text{Block: } T - F = ma, \quad F = \mu R = \mu mg; \qquad \text{Hanging mass: } Mg - T = Ma.$$

Adding eliminates T:

$$Mg - \mu mg = (M + m)a \implies a = \frac{(M - \mu m)g}{M + m}, \qquad T = M(g - a) = \frac{Mm(1 + \mu)g}{M + m}.$$

WORKED EXAMPLE 3 **Rough table: full analysis**

A block of mass 4 kg lies on a rough horizontal table with coefficient of friction $\mu = 0.25$. It is joined by a light inextensible string over a smooth edge pulley to a hanging mass of 6 kg. The system is released from rest. Take $g = 9.8\ \text{m s}^{-2}$.

1. Show that motion occurs.
2. Find the acceleration and the tension.
3. Find the speed of the hanging mass after it has fallen 0.8 m.

SOLUTION

Solution. (1) Maximum friction: $F_{\max} = \mu mg = 0.25 \times 4 \times 9.8 = 9.8$ N. Driving weight: $Mg = 6 \times 9.8 = 58.8$ N. Since $58.8 > 9.8$, motion occurs. (2) Block: $T - 9.8 = 4a$. Hanging mass: $58.8 - T = 6a$. Adding: $49 = 10a$, so $a = 4.9\ \text{m s}^{-2} > 0$ — consistent with our assumption of motion ✓. Then $T = 9.8 + 4(4.9) = 29.4$ N. (3) Using $v^2 = u^2 + 2as$ with $u = 0$, $a = 4.9$, $s = 0.8$:

$$v^2 = 2 \times 4.9 \times 0.8 = 7.84 \implies v = 2.8\ \text{m s}^{-1}.$$

WORKED EXAMPLE 4 **When the system does not move**

The block of the previous example is replaced by one of mass 10 kg, with $\mu = 0.7$, the hanging mass remaining 6 kg. Describe the motion.

SOLUTION

Solution. $F_{\max} = \mu m g = 0.7 \times 10 \times 9.8 = 68.6$ N, while the driving weight is only $Mg = 58.8$ N. Since $58.8 < 68.6$, friction can balance the pull: the system remains at rest. The actual friction is $F = 58.8$ N $< \mu R$, and the tension is $T = Mg = 58.8$ N (the hanging mass is in equilibrium). Writing $a = \dfrac{(M - \mu m)g}{M + m}$ would give a negative value — the signal that the motion assumption fails.

2.3 Using the suvat Equations

Once the (constant) acceleration is known, all the constant-acceleration formulae apply to each particle, since both share a:

$$v = u + at, \qquad s = ut + \tfrac{1}{2}at^2, \qquad v^2 = u^2 + 2as, \qquad s = \tfrac{1}{2}(u + v)t.$$

Typical examination questions ask for the time taken by the table particle to reach the pulley, or the speed of the hanging mass when it hits the floor. Remember that the two particles describe the same distance along the string while it remains taut.

WORKED EXAMPLE 5 **Time to reach the edge**

In the first worked example (smooth table, $m = 2$, $M = 3$, $a = 5.88\ \text{m s}^{-2}$), the block starts 1.5 m from the pulley. Find the time taken to reach the pulley and its speed on arrival.

SOLUTION

Solution. From $s = ut + \tfrac{1}{2}at^2$ with $u = 0$:

$$1.5 = \tfrac{1}{2}(5.88)t^2 \implies t^2 = \frac{3}{5.88} = 0.5102 \implies t \approx 0.714\ \text{s}.$$

Speed: $v = at = 5.88 \times 0.714 \approx 4.2\ \text{m s}^{-1}$ (check with $v^2 = 2as = 17.64$, $v = 4.2$ ✓).

2.4 Extension: Towing Problems with Resistance

The same logic governs a **car towing a trailer**, with the tow-bar playing the role of the string. The difference is that a tow-bar can push as well as pull: when the car brakes, the bar is in **compression** and exerts a **thrust**. Resistance forces (air resistance, rolling resistance) act on each vehicle, constant or proportional to speed; at GCE level they are usually given as constants.

METHOD **Car and trailer**

1. Draw separate force diagrams for the car and the trailer; mark driving force, resistances, and the force T in the tow-bar. Assume T is a tension (pull) — i.e. the bar pulls the trailer forward and the car backward.
2. Write $F = ma$ for each vehicle along the direction of motion (take the direction of motion as positive).
3. Add the equations to get a for the whole system, then back-substitute into one equation to find T.
4. If T comes out negative, the bar is in thrust (compression) of magnitude $|T|$ — typical under braking.

WORKED EXAMPLE 6 **Braking car and trailer: thrust in the tow-bar**

A car of mass 1200 kg tows a trailer of mass 400 kg along a horizontal road. The car brakes, producing a braking force of 6000 N; resistances of 200 N on the car and 100 N on the trailer also act. Find the deceleration and the force in the tow-bar, stating whether it is a tension or a thrust.

SOLUTION

Solution. Take the direction of motion as positive. All resistive forces are negative. Whole system:

$$-(6000+200+100) = (1200+400)a \implies -6300 = 1600a \implies a = -\frac{6300}{1600} = -3.9375\ \text{m s}^{-2}.$$

The deceleration is $3.9375\ \text{m s}^{-2}$. Now consider the trailer alone. Assume the tow-bar is in tension, so it pulls the trailer forward with force T (positive). The resistance on the trailer is 100 N backward.

$$T - 100 = 400 \times (-3.9375) = -1575 \implies T = -1475\ \text{N}.$$

Since T is negative, our assumption of tension is wrong; the bar is in compression (thrust) of magnitude 1475 N. The bar pushes the trailer backward (and the car forward) to slow the trailer down.

WARNING **Sign conventions in towing**

Decide at the start which way is positive and assume the tow-bar force is a tension (pull). Keep that convention for both vehicles: the bar exerts equal and opposite forces on car

and trailer (Newton's third law). A negative result for T means the bar is in thrust. Mixing conventions is the most common source of sign errors in braking questions.

DID YOU KNOW? **Friction in daily life**

The friction model $F = \mu R$ is an approximation (Amontons–Coulomb law). Real friction depends on contact area, temperature, and speed. In Cameroon, think of a taxi braking on a wet road in Douala: μ drops dramatically, increasing stopping distance. Engineers design brake systems to maximise μ while avoiding wheel lock (which switches from static to kinetic friction, reducing μ). This is why anti-lock braking systems (ABS) are vital for safety.

KEY POINTS **Key points of Section 2**

- Smooth table with hanging mass M driving mass m: $a = \dfrac{Mg}{M+m}$, $T = \dfrac{Mmg}{M+m}$.
- Rough table: first test Mg against μmg. Motion occurs only if $Mg > \mu mg$; then $a = \dfrac{(M-\mu m)g}{M+m}$ and $F = \mu R$ opposes the motion.
- On a horizontal surface with no vertical acceleration, $R = mg$.
- The whole-system equation yields a only; tension requires a separate single-particle equation.
- With constant a found, use the suvat formulae for times, speeds and distances.
- In towing problems, assume tension in the tow-bar; a negative result signals a thrust (compression), typical under braking.

EXERCISE 2 **Practice**

1. A mass of 5 kg on a smooth table is connected over a smooth edge pulley to a hanging mass of 3 kg. Find the acceleration, the tension, and the speed of the system after 0.5 s from rest. ($g = 9.8\ \text{m s}^{-2}$)
2. A block of mass 8 kg rests on a rough horizontal table, $\mu = 0.3$, connected to a hanging mass of 2 kg. Determine whether the system moves; if it does, find a and T; if not, find the friction and the tension.
3. Repeat question 2 with a hanging mass of 5 kg, and find how long the block takes to travel 1.2 m from rest.

4. A car of mass 1000 kg towing a trailer of mass 250 kg accelerates with driving force 3200 N against total resistances of 450 N on the car and 150 N on the trailer. Find the acceleration and the tension in the tow-bar.

SOLUTION **Exercise 1**

$a = \dfrac{3g}{8} = 3.675\ \text{m s}^{-2}$; $T = 5a = 18.375\ \text{N}$; $v = at = 3.675 \times 0.5 \approx 1.84\ \text{m s}^{-1}$.

SOLUTION **Exercise 2**

$F_{\max} = 0.3 \times 8 \times 9.8 = 23.52\ \text{N}$; driving weight $= 2g = 19.6\ \text{N} < 23.52\ \text{N}$. No motion: $a = 0$, $F = 19.6\ \text{N}$, $T = 19.6\ \text{N}$.

SOLUTION **Exercise 3**

$5g = 49 > 23.52$, so motion occurs. $a = \dfrac{(5 - 0.3 \times 8) \times 9.8}{13} = \dfrac{2.6 \times 9.8}{13} = 1.96\ \text{m s}^{-2}$; $T = 5(g - a) = 5 \times 7.84 = 39.2\ \text{N}$. From $s = \tfrac{1}{2}at^2$: $t = \sqrt{\dfrac{2 \times 1.2}{1.96}} = \sqrt{1.2245} \approx 1.11\ \text{s}$.

SOLUTION **Exercise 4**

Whole system: $3200 - 450 - 150 = 1250a$, so $a = \dfrac{2600}{1250} = 2.08\ \text{m s}^{-2}$. Trailer: $T - 150 = 250 \times 2.08 = 520$, hence $T = 670\ \text{N}$ (tension, as expected under acceleration).

3 SECTION 3: MOTION ON INCLINED PLANES

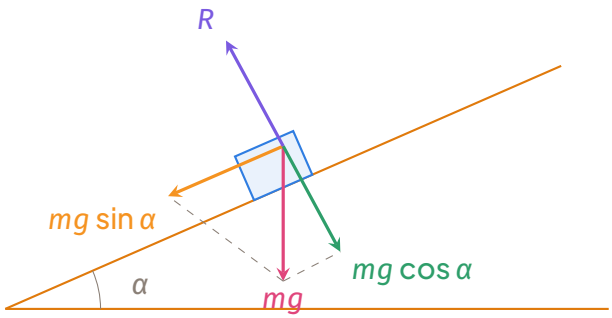
In Section 2 we studied connected particles on horizontal surfaces, where the weight of a block was completely balanced by the normal reaction ($R = mg$) and only the tension and friction acted along the direction of motion. Life in Cameroon, however, is rarely horizontal: think of a truck climbing the steep road to Dschang, a wheelbarrow pushed up a plank onto a lorry in a Douala warehouse, or a crate sliding down a ramp. On a slope, the weight itself has a component along the surface, and this changes everything: the normal reaction, the friction, and the driving force of the motion. This section teaches you to handle such problems with complete confidence.

3.1 Resolving the Weight on an Inclined Plane

Observation. Place a book on a flat table and tilt the table slowly. At first nothing happens; then, beyond a certain angle, the book slides. Gravity has not changed — what has changed is the direction in which gravity can push the book relative to the surface. To understand the motion we must split the weight into two perpendicular parts: one pulling the object along the plane, and one pressing it into the plane.

Consider a particle of mass m on a plane inclined at an angle α to the horizontal. The weight mg acts vertically downwards. We resolve it into:

- a component **parallel to the plane**, pointing down the slope;
- a component **perpendicular to the plane**, pressing into the surface.



Resolving the weight on a plane inclined at α: the component $mg\sin\alpha$ acts down the plane, $mg\cos\alpha$ presses into the plane, and the normal reaction R acts perpendicular to the surface.

The geometry is the key point. The angle between the vertical (direction of mg) and the normal to the plane is exactly α, because the normal is tilted by α from the vertical just as the plane is tilted by α from the horizontal. Hence, in the right-angled triangle formed by the weight and its two components:

$$\text{component down the plane} = mg\sin\alpha, \qquad \text{component into the plane} = mg\cos\alpha.$$

PROPERTY **Weight components and normal reaction on an inclined plane**

For a particle of mass m resting on a plane inclined at angle α to the horizontal:

- the component of weight down the plane is $mg\sin\alpha$;
- the component of weight perpendicular into the plane is $mg\cos\alpha$;
- since there is no acceleration perpendicular to the plane (the particle neither sinks into nor lifts off the surface), resolving perpendicular to the plane gives

$$R = mg\cos\alpha.$$

This result is examinable and must be justified by resolving perpendicular to the plane.

METHOD **Resolving parallel and perpendicular to the plane**

1. Draw a free-body diagram: weight mg vertically down, normal reaction R perpendicular to the plane, friction (if any) along the plane opposing motion, tension (if any) along the string.
2. Choose axes parallel and perpendicular to the plane.
3. Replace mg by its two components $mg\sin\alpha$ (down the plane) and $mg\cos\alpha$ (into the plane).
4. Write $\sum F = ma$ along the plane, and $\sum F = 0$ perpendicular to the plane.
5. Never mix horizontal/vertical components with parallel/perpendicular components in the same equation. Choose one pair of axes and stay with it.

WARNING **The most common mistake on inclined planes**

Many candidates write $R = mg$ out of habit from horizontal problems. On an inclined plane this is wrong: the correct reaction is

$$R = mg\cos\alpha.$$

Since friction is $F = \mu R$, using $R = mg$ also corrupts the friction force and destroys the whole solution. Always recompute R by resolving perpendicular to the plane.

3.2 A Single Particle Sliding on an Inclined Plane

Smooth plane. With no friction, the only force along the plane is the component $mg\sin\alpha$. Newton's second law down the plane gives

$$mg\sin\alpha = ma \quad\Longrightarrow\quad \boxed{a = g\sin\alpha.}$$

Notice that the acceleration is independent of the mass — just as in free fall, but reduced by the factor $\sin\alpha$.

Rough plane, particle sliding down. Friction now acts up the plane (opposing the motion), with limiting value $F = \mu R = \mu mg\cos\alpha$. Newton's second law down the plane:

$$mg\sin\alpha - \mu mg\cos\alpha = ma \quad\Longrightarrow\quad \boxed{a = g(\sin\alpha - \mu\cos\alpha).}$$

When does the particle stay at rest? For the particle to remain in equilibrium, friction must be able to balance the driving component: we need $mg\sin\alpha \le \mu mg\cos\alpha$, i.e.

$$\tan\alpha \le \mu.$$

DEFINITION **Angle of friction**

The **angle of friction** λ is defined by

$$\tan\lambda = \mu.$$

A particle placed on a rough plane remains at rest if and only if the inclination satisfies $\alpha \le \lambda$, equivalently $\tan\alpha \le \mu$. If the plane is tilted beyond λ, the particle must slide. (This is a useful GCE extra: it connects the equilibrium condition directly to the coefficient of friction.)

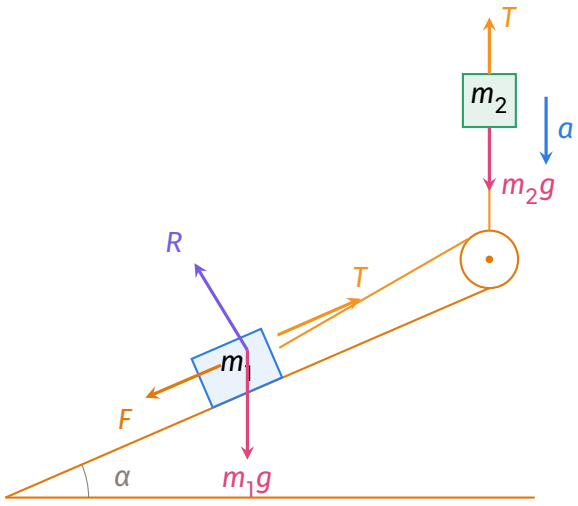
EXEMPLE **Checking equilibrium on a rough plank**

A crate of mass 25 kg rests on a plank inclined at 20° to the horizontal, with $\mu = 0.4$. Does it stay at rest? Take $g = 9.8\ \text{m s}^{-2}$.

Solution. $\tan 20^\circ \approx 0.364 < 0.4 = \mu$, so $\alpha < \lambda$ and the crate remains at rest. The friction needed is only $mg\sin 20^\circ \approx 25 \times 9.8 \times 0.342 \approx 83.8$ N, while the maximum available is $\mu mg\cos 20^\circ \approx 0.4 \times 25 \times 9.8 \times 0.940 \approx 92.1$ N. Friction is not limiting — the crate is safe.

3.3 Particle on a Plane Connected to a Hanging Particle

The classic GCE configuration: a block A of mass m_1 rests on a plane inclined at α, connected by a light inextensible string parallel to the plane, passing over a smooth pulley fixed at the top, to a particle B of mass m_2 hanging freely.



Block m_1 on a rough inclined plane connected over a smooth pulley to hanging mass m_2. Free-body elements: on the slope, T up the plane, friction F opposing motion, R perpendicular, weight resolved; on the hanging mass, T up and $m_2 g$ down.

Setting up the equations. Suppose the system moves with m_2 descending (so m_1 moves up the plane) with common acceleration a, and the plane is rough with friction on m_1 acting down the plane (opposing its motion). Then:

- Perpendicular to the plane (no acceleration): $R = m_1 g \cos\alpha$.
- Along the plane, for m_1 (taking up the plane as positive):

$$T - m_1 g \sin\alpha - \mu m_1 g \cos\alpha = m_1 a. \tag{1}$$

- Vertically, for the hanging mass m_2 (taking downwards as positive):

$$m_2 g - T = m_2 a. \tag{2}$$

Adding (1) and (2) eliminates T:

$$a = \frac{m_2 g - m_1 g \sin\alpha - \mu m_1 g \cos\alpha}{m_1 + m_2}, \qquad T = m_2(g - a).$$

WARNING **Assuming the direction of motion**

When the direction of motion is not obvious, assume one, write the equations consistently (friction opposing the assumed motion), and solve. If the computed a is negative, the actual motion is in the opposite direction — but beware: you must then redo the problem with friction reversed, because friction always opposes the real motion. If the driving force is smaller than the maximum available friction plus the opposing weight component, the system may simply not move at all; a negative acceleration under a limiting-friction assumption can mean equilibrium, not reversed motion.

WORKED EXAMPLE 7 **Rough plane with hanging mass: finding a and T**

A block A of mass 4 kg lies on a rough plane inclined at 30° to the horizontal, with coefficient of friction $\mu = 0.2$. It is connected by a light inextensible string, parallel to the plane and passing over a smooth pulley at the top, to a particle B of mass 6 kg hanging freely. The system is released from rest. Take $g = 9.8\ \text{m s}^{-2}$. Find the acceleration of the system and the tension in the string, and check consistency.

SOLUTION

Step 1 — Guess the direction. Driving force if B descends: $m_2 g = 6 \times 9.8 = 58.8$ N. Resisting: component of A's weight down the plane, $m_1 g \sin 30^\circ = 4 \times 9.8 \times 0.5 = 19.6$ N, plus friction $\mu m_1 g \cos 30^\circ = 0.2 \times 4 \times 9.8 \times 0.866 \approx 6.79$ N. Total resistance ≈ 26.4 N < 58.8 N, so B does descend and A moves up the plane.
Step 2 — Normal reaction. Perpendicular to the plane:

$$R = m_1 g \cos 30^\circ = 4 \times 9.8 \times 0.866 \approx 33.95\ \text{N}.$$

Step 3 — Equations of motion. For A up the plane, with friction $F = \mu R \approx 6.79$ N down the plane:

$$T - 19.6 - 6.79 = 4a. \tag{1}$$

For B descending:

$$58.8 - T = 6a. \tag{2}$$

Step 4 — Solve. Adding (1) and (2):

$$58.8 - 26.39 = 10a \quad \Longrightarrow \quad a = \frac{32.41}{10} \approx 3.24\ \text{m s}^{-2}.$$

From (2): $T = 58.8 - 6(3.24) \approx 58.8 - 19.45 \approx 39.4$ N.
Step 5 — Consistency check. $a > 0$, confirming our assumed direction. Also $T = 39.4$ N lies between $m_1 g \sin\alpha + F \approx 26.4$ N and $m_2 g = 58.8$ N, exactly as expected: the tension must both drag A up and brake B's fall. $\boxed{a \approx 3.24\ \text{m s}^{-2},\ T \approx 39.4\ \text{N}}$

WORKED EXAMPLE 8 **Smooth plane and suvat follow-up: speed at the bottom**

A particle of mass 2 kg is released from rest at the top of a smooth plane of length 5 m inclined at 30°. Take $g = 9.8\ \text{m s}^{-2}$. Find its acceleration and its speed at the bottom of the plane.

SOLUTION

Acceleration. On a smooth plane, $a = g\sin\alpha = 9.8 \times \sin 30^\circ = 4.9\ \text{m s}^{-2}$ down the plane.
Speed at the bottom. Using $v^2 = u^2 + 2as$ with $u = 0$, $a = 4.9$, $s = 5$:

$$v^2 = 0 + 2 \times 4.9 \times 5 = 49 \quad \Longrightarrow \quad v = 7\ \text{m s}^{-1}.$$

Note the elegant check: $v^2 = 2g\sin\alpha \cdot s = 2gh$, where $h = s\sin\alpha$ is the vertical drop — exactly the free-fall result, foreshadowing the energy methods of Section 6.

3.4 When the String Goes Slack: Motion Up the Plane

A favourite GCE twist: the hanging mass hits the ground (or the string breaks) while the block on the plane is still moving up. From that instant the tension vanishes, and the block decelerates under the component of its own weight (and friction, if rough) until it stops momentarily.

WORKED EXAMPLE 9 **Distance travelled up the plane after the string goes slack**

In the worked example above (rough plane, $\mu = 0.2$, $\alpha = 30^\circ$, $m_1 = 4$ kg), suppose the string breaks when block A is moving up the plane at $3\ \text{m s}^{-1}$. How much further does A travel up the plane before coming to instantaneous rest? Take $g = 9.8\ \text{m s}^{-2}$.

SOLUTION

New equation of motion. With the string slack, the forces on A along the plane are the weight component $m_1 g \sin\alpha$ and friction $\mu m_1 g \cos\alpha$, both acting down the plane (friction opposes the upward motion):

$$-m_1 g \sin\alpha - \mu m_1 g \cos\alpha = m_1 a \quad \Longrightarrow \quad a = -g(\sin\alpha + \mu\cos\alpha).$$

Numerically: $a = -9.8(0.5 + 0.2 \times 0.866) = -9.8 \times 0.6732 \approx -6.60\ \text{m s}^{-2}$.
Suvat. With $u = 3$, $v = 0$, $a = -6.60$:

$$v^2 = u^2 + 2as \implies 0 = 9 - 2 \times 6.60 \times s \implies s = \frac{9}{13.2} \approx 0.68\ \text{m}.$$

The block travels about 0.68 m further up before stopping. (Will it then slide back? Yes, since $\tan 30^\circ \approx 0.577 > \mu = 0.2$: the rest condition fails, so it accelerates back down with $a = g(\sin\alpha - \mu\cos\alpha) \approx 3.20\ \text{m s}^{-2}$.)

KEY POINTS **Key points of Section 3**

- On a plane inclined at α: weight splits into $mg\sin\alpha$ down the plane and $mg\cos\alpha$ into the plane; hence $R = mg\cos\alpha$, never mg.
- Smooth plane: $a = g\sin\alpha$. Rough plane sliding down: $a = g(\sin\alpha - \mu\cos\alpha)$.
- Rest is possible if and only if $\tan\alpha \le \mu$, i.e. $\alpha \le \lambda$ where $\tan\lambda = \mu$ is the angle of friction.
- For a block on a plane connected to a hanging mass: write one equation along the plane for the block, one vertical equation for the hanging mass, then add to eliminate T.
- If the assumed direction of motion gives $a < 0$, reverse the direction and the friction, and solve again — or conclude the system stays at rest.
- After the string goes slack, recompute the acceleration with $T = 0$ before applying suvat.

EXERCISE 3 **Practice**

A block of mass 5 kg rests on a rough plane inclined at 25° to the horizontal, with $\mu = 0.3$. It is connected by a light inextensible string parallel to the plane, over a smooth pulley at the top, to a hanging particle of mass 3 kg. Take $g = 9.8\ \text{m s}^{-2}$.

1. Show that the system does move, and state which way.

2. Find the acceleration and the tension in the string.
3. The hanging mass hits the ground after descending 1.5 m. Find the speed of the block at that instant.
4. After the string goes slack, find the further distance the block moves up the plane before momentary rest.

SOLUTION **Exercise 1**

1. Driving force (hanging side): $3 \times 9.8 = 29.4$ N down. Resisting if A moves up: $5 \times 9.8 \times \sin 25^\circ \approx 20.7$ N plus maximum friction $0.3 \times 5 \times 9.8 \times \cos 25^\circ \approx 13.3$ N, total ≈ 34.0 N > 29.4 N. Check the other direction: driving force down the plane is 20.7 N against 29.4 N (weight of B) plus friction up the plane (max 13.3 N) — impossible. Since $20.7 + 13.3 = 34.0 > 29.4 > 20.7 - 13.3 = 7.4$, friction can hold the system: it does not move (static equilibrium, friction $= 29.4 - 20.7 = 8.7$ N up the plane, below the limiting 13.3 N).
2–4. Because the system remains at rest, parts 2, 3 and 4 have no physical solution with the given data. If the plane were smoother (e.g. $\mu = 0.1$), then motion would occur with the 3 kg mass descending: $a = \dfrac{29.4 - 20.7 - 4.44}{8} \approx 0.53\ \text{m s}^{-2}$, $T = 3(9.8 - 0.53) \approx 27.8$ N; after 1.5 m, $v = \sqrt{2 \times 0.53 \times 1.5} \approx 1.26\ \text{m s}^{-1}$; after the string goes slack, deceleration $g(\sin 25^\circ + 0.1 \cos 25^\circ) \approx 5.03\ \text{m s}^{-2}$, giving further distance $s = \dfrac{1.26^2}{2 \times 5.03} \approx 0.16$ m. The essential lesson: always test whether the system moves before computing an acceleration.

DID YOU KNOW? **Inclined planes in daily life and in history**

Galileo Galilei used inclined planes precisely because $a = g \sin \alpha$ is a diluted version of free fall: with the crude clocks of his time, a ball rolling slowly down a gentle slope allowed him to measure that distances grow as the square of time — the birth of kinematics. Closer to home, every ramp at a Douala building site, every switchback on the Bamenda escarpment road, is an applied lesson in $mg \sin \alpha$: engineers choose the gradient so that vehicles can climb without the driving wheels losing grip, i.e. so that the required friction stays below $\mu R = \mu mg \cos \alpha$.

4 SECTION 4: ATWOOD'S MACHINE AND MULTI-STAGE MOTION

In Section 3 we analysed particles connected across smooth and rough inclined planes. The

core strategy was to write Newton's second law separately for each particle, then couple the equations through the common tension and the common magnitude of acceleration. We now apply this same rigorous method to the classic **Atwood's machine**: two masses hanging vertically from a light inextensible string passing over a smooth fixed pulley. This system is a favourite of GCE examiners because it yields clean algebra, illustrates the string constraint perfectly, and leads naturally to multi-stage problems where the forces change abruptly during the motion.

4.1 The Atwood's Machine: Setting Up the Model

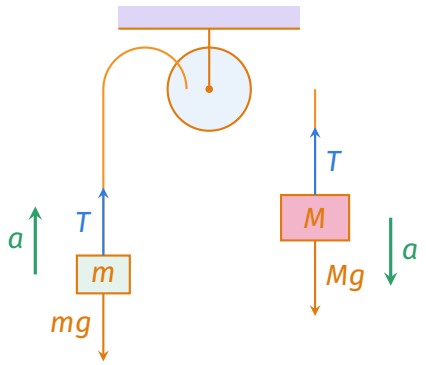
EXPERIMENT **Observing an Atwood's machine**

Hang a light string over a smooth fixed pulley (a well-oiled bicycle wheel works well in the school laboratory) and attach a mass of 200 g to one end and 100 g to the other. Release the system from rest. The heavier mass descends, the lighter one rises, and — this is the key observation — both move slowly, much more slowly than a freely falling stone. The system acts as a "diluted gravity" device: by choosing the masses appropriately, we can make the acceleration as small as we wish. This is precisely why George Atwood invented it in 1784: to study accelerated motion with the crude timing devices available at the time.

DEFINITION **Atwood's machine**

An **Atwood's machine** consists of two particles of masses m and M (with $M > m$) attached to the ends of a light inextensible string passing over a smooth fixed pulley, and released from rest. The modelling assumptions have precise consequences:

- Light string: its mass is negligible, so the tension T is the same throughout the string.
- Smooth pulley: no friction at the axle, so the tension is the same on both sides of the pulley.
- Inextensible string: its length does not change, therefore the two particles always have accelerations of the same magnitude a, but in opposite vertical directions (one up, one down).



Atwood's machine: forces on each mass and the common magnitude a of the accelerations.

Equations of motion. Since $M > m$, mass M accelerates downwards and mass m accelerates upwards. The most efficient sign convention is to choose the positive direction for each particle separately to be its actual direction of motion. This automatically respects the string constraint (equal magnitude a) and avoids sign errors.

- For M (positive direction **downwards**): the resultant force is $Mg - T$. Newton's second law gives

$$Mg - T = Ma. \tag{1}$$

- For m (positive direction **upwards**): the resultant force is $T - mg$. Newton's second law gives

$$T - mg = ma. \tag{2}$$

Adding equations (1) and (2) eliminates the unknown tension T:

$$Mg - mg = Ma + ma \quad \Longrightarrow \quad (M - m)g = (M + m)a.$$

PROPERTY **Standard results for the Atwood's machine**

For an Atwood's machine with $M > m$, released from rest:

$$a = \frac{(M - m)g}{M + m}, \qquad T = \frac{2Mmg}{M + m}.$$

The tension always satisfies $mg < T < Mg$: it exceeds the weight of the rising mass (it must pull it up) but is less than the weight of the falling mass (which must have a net downward force). The derivation above is examinable — do not quote the formulae without justification unless the question explicitly says "you may assume".

EXEMPLE **Quick check of the bounds**

Take $M = 5\,\text{kg}$, $m = 3\,\text{kg}$, $g = 9.8\,\text{m s}^{-2}$. Then

$$a = \frac{(5-3)\times 9.8}{5+3} = \frac{2\times 9.8}{8} = 2.45\,\text{m s}^{-2}, \qquad T = \frac{2\times 5\times 3\times 9.8}{8} = 36.75\,\text{N}.$$

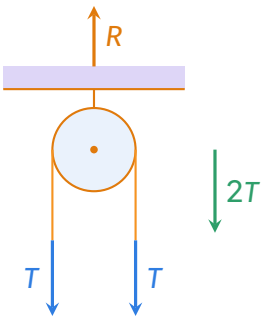
Check: $mg = 29.4\,\text{N} < 36.75\,\text{N} < 49\,\text{N} = Mg$. ✓

WARNING **Sign errors with opposite directions**

The two masses move in opposite directions. A classic error is to write both equations with a single global "up positive" convention, obtaining $Mg - T = Ma$ and $T - mg = -ma$, and then adding to get $Mg - mg = 0$ (nonsense). Fix the positive direction for each particle separately, in its own direction of motion, and state it clearly on your diagram. The string constraint then simply reads: accelerations equal in magnitude, opposite in direction.

4.2 Force on the Pulley Support

The pulley has two strands of string pulling down on it, each with tension T. Since the pulley is smooth and light, it merely redirects the tensions; it does not alter their magnitude.



Support reaction $R = 2T$ (light pulley)

The pulley support carries the downward force $2T$ from the two strands of the string.

METHOD **Force on the pulley:** $R = 2T$

1. The pulley feels two downward pulls of magnitude T, one from each vertical strand of the string. The total downward force on the pulley is therefore $2T$.
2. For a light pulley in vertical equilibrium, the upward reaction force R from the support equals this total downward force: $R = 2T$.

3. Compare R with the total weight $(M+m)g$. Using $T = \dfrac{2Mmg}{M+m}$,

$$R = 2T = \frac{4Mmg}{M+m}.$$

The difference between the total weight and the support reaction is

$$(M+m)g - R = (M+m)g - \frac{4Mmg}{M+m} = \frac{(M+m)^2 g - 4Mmg}{M+m} = \frac{(M-m)^2 g}{M+m}.$$

Notice that $(M-m)g = (M+m)a$, so $(M-m)^2 g/(M+m) = (M-m)a$. Hence

$$R = (M+m)g - (M-m)a.$$

4. The support carries less than the combined weight $(M+m)g$. The "missing weight" is $(M-m)a$, which is exactly the net force acting on the two-particle system (mass M accelerating down, mass m accelerating up).

DID YOU KNOW? **Weighing an accelerating system**

If the whole Atwood's machine (pulley, string, and masses) were placed on a very sensitive scale, the scale would read $R = 2T$ (plus the pulley's own weight if not light), not $(M+m)g$. The same principle explains why a lift carrying accelerating passengers exerts a different force on its cable than the static weight — a theme we revisit in real-life problems later in this chapter.

4.3 Kinematics of the Machine

Because the acceleration a is constant, all the suvat equations apply to each particle individually. The string constraint guarantees that at every instant:

- the two masses have **equal speeds** (but opposite directions);
- the distance descended by M equals the distance ascended by m.

WORKED EXAMPLE 10 **Basic Atwood's machine kinematics**

Two particles of masses 2 kg and 5 kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is released from rest with the 5 kg mass 1.4 m above the ground. Take $g = 9.8\ \text{m s}^{-2}$. Find (a) the acceleration, (b) the tension, (c) the speed with which the 5 kg mass hits the ground, (d) the time taken.

SOLUTION

Solution. Let $M = 5$ kg (moving down), $m = 2$ kg (moving up). Positive direction: down for M, up for m.

$$5g - T = 5a \quad (1), \qquad T - 2g = 2a \quad (2).$$

(a) Adding (1) and (2): $3g = 7a$, so $a = \dfrac{3 \times 9.8}{7} = 4.2\,\text{m s}^{-2}$. (b) From (2): $T = 2g + 2a = 19.6 + 8.4 = 28$ N. (c) For the 5 kg mass: $u = 0$, $s = 1.4$ m, $a = 4.2\,\text{m s}^{-2}$. Using $v^2 = u^2 + 2as$:

$$v^2 = 2 \times 4.2 \times 1.4 = 11.76 \quad \Longrightarrow \quad v = \sqrt{11.76} = 3.43\,\text{m s}^{-1}\ (3\text{ s.f.}).$$

(d) Using $v = u + at$: $t = \dfrac{3.43}{4.2} = 0.817$ s (3 s.f.).

4.4 Multi-Stage Motion: When the String Goes Slack

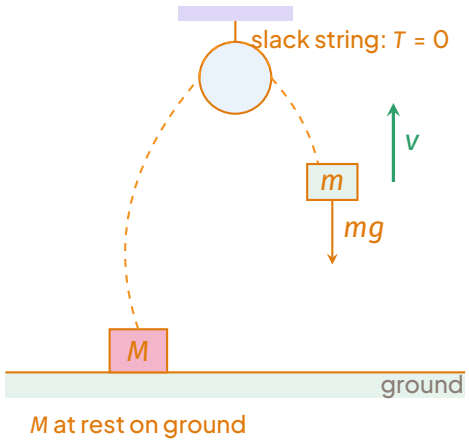
GCE questions often involve a scenario where the heavier mass starts above the ground and the lighter mass starts on (or near) the ground. Three distinct stages occur:

Stage 1: Connected motion. Both masses move together with acceleration $a = \dfrac{(M - m)g}{M + m}$ until M hits the ground (or the string breaks). At that instant both masses have the same speed v.

Stage 2: Free projectile motion (string slack). Once M lands, the string goes slack: the tension drops instantly to zero. Mass m is now moving upwards with speed v, acted on by gravity alone. It continues to rise, decelerating at g, until it comes momentarily to rest. Using $v^2 = u^2 + 2as$ with final speed 0 and acceleration $-g$ (taking upward as positive), the extra height gained is

$$h = \frac{v^2}{2g}, \qquad \text{time to momentary rest: } t = \frac{v}{g}.$$

Stage 3: The string becomes taut again (qualitative only). After reaching its highest point, m falls back down. The slack string gradually straightens, and when it becomes taut there is a sudden jerk: an impulsive tension that almost instantaneously changes the velocities of both particles (and may set M moving again if it can leave the ground). The analysis of this jerk requires impulse and momentum, which is treated in a later chapter — at GCE you will only ever be asked for a qualitative comment here.



Stage 2: M has landed, the string is slack, and m rises freely under gravity alone.

METHOD **Solving multi-stage problems**

1. Draw a separate force diagram for each stage. Forces change between stages — never recycle a diagram.
2. Stage 1 (connected): Apply Newton's second law to each particle, solve for a and T, then use suvat to find the speed v at the moment the stage ends (when a mass hits the ground or the string breaks).
3. Carry over the velocity: The final velocity of one stage is the initial velocity of the next. Both masses have the same speed v at the transition.
4. Stage 2 (slack string): The free particle moves under gravity alone ($a = \pm g$). Use $v^2 = u^2 - 2gh$ for the extra height and $t = u/g$ for the time to momentary rest.
5. Assemble total times and total heights from the stages, and comment qualitatively on any jerk if asked.

WARNING **Do not keep the tension after the string goes slack**

The single most common error in these questions is to continue using $T = \dfrac{2Mmg}{M+m}$ — or indeed any tension at all — after M has landed. Once the string is slack, $T = 0$ and the rising mass is a projectile with acceleration g downwards. Its deceleration is g, not the Stage-1 value a. Similarly, never use the Stage-1 acceleration to compute the extra height or the time to rest.

WORKED EXAMPLE 11 **GCE-style multi-stage problem**

Particles P and Q, of masses 0.3 kg and 0.5 kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed pulley. The system is released from rest with both particles 0.8 m above horizontal ground. Take $g = 9.8\ \text{m s}^{-2}$. (a) Find the acceleration of the system and the tension in the string. (b) Find the speed of Q when it reaches the ground. (c) After Q hits the ground (and does not rebound), find the greatest height above the ground reached by P. (d) Find the total time from release until P reaches its greatest height. (e) State, with a reason, what happens to the tension in the string at the instant Q lands.

SOLUTION

Solution. (a) Let $M = 0.5$ kg (down positive) and $m = 0.3$ kg (up positive).

$$0.5g - T = 0.5a \quad (1), \qquad T - 0.3g = 0.3a \quad (2).$$

Adding: $0.2g = 0.8a$, so $a = \dfrac{0.2 \times 9.8}{0.8} = 2.45\ \text{m s}^{-2}$. From (2): $T = 0.3(g + a) = 0.3 \times 12.25 = 3.675$ N.

(b) Stage 1 ends when Q has fallen 0.8 m. With $u = 0$:

$$v^2 = 2 \times 2.45 \times 0.8 = 3.92 \quad \Longrightarrow \quad v = \sqrt{3.92} = 1.98\ \text{m s}^{-1}\ (3\ \text{s.f.}).$$

At this instant P is $0.8 + 0.8 = 1.6$ m above the ground, moving up at $1.98\ \text{m s}^{-1}$.

(c) Stage 2: string slack, P rises under gravity alone. Extra height:

$$h = \frac{v^2}{2g} = \frac{3.92}{19.6} = 0.2\ \text{m}.$$

Greatest height above ground $= 1.6 + 0.2 =$ **1.8 m**.

(d) Stage 1 time: $t_1 = \dfrac{v}{a} = \dfrac{1.98}{2.45} = 0.808$ s. Stage 2 time: $t_2 = \dfrac{v}{g} = \dfrac{1.98}{9.8} = 0.202$ s. Total time $= t_1 + t_2 = 1.01$ s (3 s.f.).

(e) When Q lands, the string goes slack and the tension falls instantaneously to zero, because there is no longer anything pulling the string taut on Q's side. P then moves under gravity alone until the string becomes taut again, at which point a jerk (impulsive tension) occurs — its analysis requires impulse and momentum, which is beyond this chapter.

EXERCISE 4 **Practice: full multi-stage problem**

Two particles of masses 1 kg and 3 kg hang from a light inextensible string over a smooth fixed pulley. The system is released from rest with the 3 kg mass 2 m above the ground and the 1 kg mass on the ground. Take $g = 9.8\ \text{m s}^{-2}$. (a) Show that the acceleration is $4.9\ \text{m s}^{-2}$ and find the tension. (b) Find the reaction at the pulley support while the system is moving, and verify that $R = (M + m)g - (M - m)a$. (c) Find the speed of the 3 kg mass as it hits the

ground. (d) Find the greatest height reached by the 1 kg mass and the total time taken to reach it.

SOLUTION **Exercise 1**

(a) Let $M = 3$ kg (down positive), $m = 1$ kg (up positive).

$$3g - T = 3a \quad (1), \qquad T - g = a \quad (2).$$

Adding: $2g = 4a \implies a = \frac{g}{2} = 4.9\ \text{m s}^{-2}$. From (2): $T = g + a = 9.8 + 4.9 = 14.7$ N.
(b) $R = 2T = 29.4$ N. Check: $(M + m)g - (M - m)a = 4 \times 9.8 - 2 \times 4.9 = 39.2 - 9.8 = 29.4\ \text{N} = R$. ✓
(c) $v^2 = 2as = 2 \times 4.9 \times 2 = 19.6$, so $v = \sqrt{19.6} = 4.43\ \text{m s}^{-1}$ (3 s.f.).
(d) At the moment M hits the ground, m is 2 m above the ground (it has risen the same distance M fell), moving up at $4.43\ \text{m s}^{-1}$. Extra height: $h = \frac{v^2}{2g} = \frac{19.6}{19.6} = 1$ m. Greatest height $= 2 + 1 =$ **3 m**. Times: $t_1 = \frac{v}{a} = \frac{4.43}{4.9} = 0.904$ s, $t_2 = \frac{v}{g} = \frac{4.43}{9.8} = 0.452$ s. Total time = 1.36 s (3 s.f.).

KEY POINTS **Key points of Section 4**

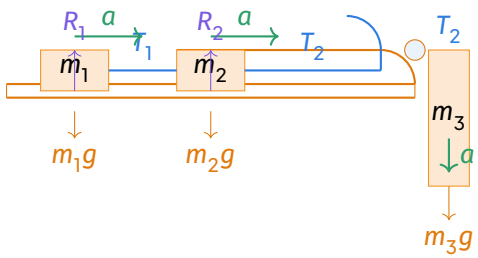
- Atwood's machine ($M > m$): $Mg - T = Ma$, $T - mg = ma$, giving $a = \frac{(M - m)g}{M + m}$ and $T = \frac{2Mmg}{M + m}$, with $mg < T < Mg$.
- The pulley support carries $R = 2T$, which is less than $(M + m)g$; the deficit is $(M - m)a$, the net force accelerating the system.
- The two masses always have equal speeds in opposite directions; distances risen and fallen are equal.
- Multi-stage motion: when a mass lands or the string breaks, the string goes slack ($T = 0$) and the free mass continues as a projectile under gravity alone; the final velocity of each stage seeds the next.
- The jerk when the string re-tightens involves impulsive tension — qualitative comment only at this stage.

5 SECTION 5: PULLEY SYSTEMS WITH MORE THAN TWO MASSES AND VARIABLE ACCELERATION

In Section 4 we mastered Atwood's machine and multi-stage motion where the acceleration was constant. We now extend our toolkit to systems with **three or more particles** and to situations where the **driving force changes with time or position**, leading to variable acceleration. The fundamental principles remain Newton's second law and the geometric constraints imposed by inextensible strings, but the algebra becomes richer and the integration techniques more varied.

5.1 Systems of Three or More Particles on Horizontal Surfaces

Consider a light inextensible string passing over a smooth pulley at the edge of a smooth horizontal table. Two particles of masses m_1 and m_2 lie on the table, connected in series, and a third particle of mass m_3 hangs vertically. The string is taut throughout. Because the string is inextensible, all three particles move with the **same magnitude of acceleration** a. However, the string is now divided into **two distinct segments**: one between m_1 and m_2 (tension T_1) and another between m_2 and the pulley (tension T_2). The hanging mass feels tension T_2 as well (same segment).



Three-mass system: two masses on a smooth table linked in series to a hanging mass. Note the two different tensions T_1 and T_2.

METHOD **One unknown tension per string segment**

When a system contains several particles connected by strings, each continuous segment of string that is separated by a particle or a pulley generally carries a different tension. Label them $T_1, T_2, \ldots$ and write Newton's second law for each particle individually. The constraint of equal acceleration magnitude (for particles on the same taut string) links the equations.

WORKED EXAMPLE 12 **Three masses on a smooth table and a hanging mass**

A particle A of mass 2 kg and a particle B of mass 3 kg rest on a smooth horizontal table,

connected by a light inextensible string. The string passes over a smooth pulley at the edge of the table and carries a particle C of mass 5 kg hanging freely. The system is released from rest. Find the acceleration of the system and the tensions T_1 (between A and B) and T_2 (between B and C).

SOLUTION

Solution. Let the acceleration be a (to the right for A and B, downward for C).

- For A (only horizontal force is T_1): $T_1 = 2a$. (1)
- For B (horizontal forces: T_2 to the right, T_1 to the left): $T_2 - T_1 = 3a$. (2)
- For C (vertical forces: weight $5g$ down, tension T_2 up): $5g - T_2 = 5a$. (3)

Add (1), (2) and (3): $5g = (2 + 3 + 5)a = 10a \Rightarrow a = \frac{g}{2} = 4.9\,\text{m s}^{-2}$. Substitute into (1): $T_1 = 2 \times 4.9 = 9.8$ N. Substitute into (3): $T_2 = 5g - 5a = 5(g - a) = 5(9.8 - 4.9) = 24.5$ N. Check: Equation (2) gives $24.5 - 9.8 = 14.7 = 3 \times 4.9$. Consistent.

PROPERTY **Whole-system equation for connected particles**

For any system of n particles connected by light inextensible strings passing over smooth pulleys, if all particles move with the same magnitude of acceleration a, then

$$a = \frac{\text{sum of external driving forces in the direction of motion}}{\text{total mass of the system}}.$$

Internal tensions cancel out when the equations are summed. This gives a quick way to find a before solving for individual tensions.

5.2 Movable Pulleys (Simple Cases)

A **movable pulley** is a pulley that can accelerate. Consider a light pulley of negligible mass carrying a particle of mass M, with a light inextensible string passing over it. One end of the string is fixed, the other is pulled with acceleration a_1, while the pulley (and mass M) accelerates with a_2. The geometry of the string gives a constraint: if the string length is constant, then $2a_2 = a_1$ (or more generally $2a_{\text{pulley}} = a_1 + a_2$ for two free ends). In the GCE Advanced Level syllabus, only simple configurations where the constraint is linear are examined.

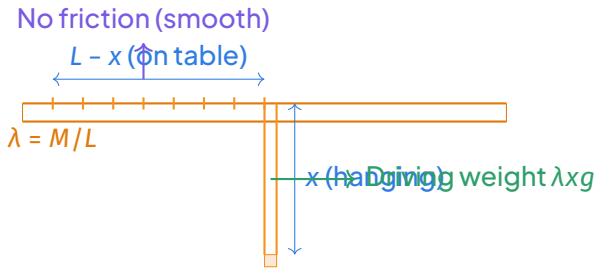
EXEMPLE **Mass on a movable pulley**

A light smooth pulley carries a mass $M = 4$ kg. A light inextensible string passes over the pulley; one end is attached to a fixed support, the other end is pulled vertically upward with a force $F = 60$ N. Find the acceleration of the mass M. Solution. Let a be the upward acceleration of the mass M (and the pulley). The free end of the string has acceleration $2a$

upward (constraint: if the pulley rises by y, the free end must be pulled up by $2y$ to keep the string length constant). The tension T is the same throughout the string (light pulley, smooth). For the mass M: $2T - Mg = Ma$. For the free end (massless): $F - T = 0 \Rightarrow T = F = 60$ N. Then $2(60) - 4g = 4a \Rightarrow 120 - 39.2 = 4a \Rightarrow a = 20.2\ \text{m s}^{-2}$ upward.

5.3 Variable Acceleration: Chain Sliding off a Table

Many real systems involve a driving force that changes as the configuration changes. A classic example is a **uniform chain** initially coiled at the edge of a smooth horizontal table, with a small length x_0 hanging over the edge. As the chain slides, the hanging length x increases, so the driving weight $\lambda x g$ (where λ is the mass per unit length) increases. The acceleration is no longer constant.

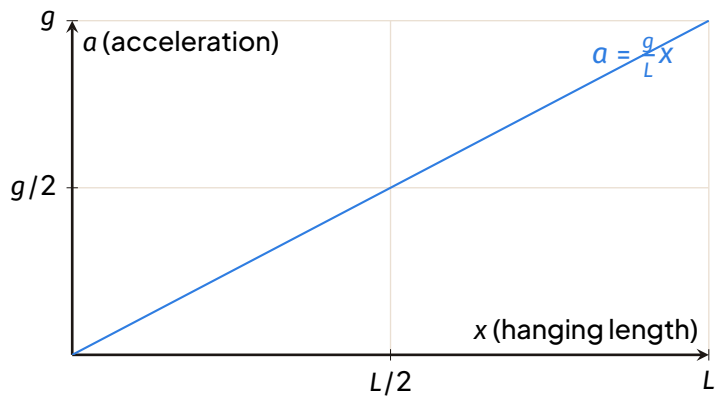


Uniform chain of length L and mass M sliding off a smooth table. The hanging length x varies with time.

Let the total mass be M, total length L, so linear density $\lambda = M/L$. At a given instant, hanging length = x, mass on table = $\lambda(L - x)$, hanging mass = λx. The only horizontal external force on the whole chain is the weight of the hanging part (the table is smooth, so no friction). Applying the whole-system equation:

$$a = \frac{\text{net driving force}}{\text{total mass}} = \frac{\lambda x g}{M} = \frac{\lambda x g}{\lambda L} = \frac{g}{L}x.$$

Thus acceleration is **proportional to the hanging length** x: $a = \frac{g}{L}x$. This is a variable acceleration problem where a is a function of displacement x.



Acceleration a as a linear function of hanging length x for a uniform chain on a smooth table.

METHOD **Solving variable acceleration problems**

When acceleration is not constant, the suvat equations do not apply. Instead:

1. Express acceleration as a function of the relevant variable: $a = f(x)$ or $a = f(t)$.
2. If $a = f(x)$: use $a = v\frac{dv}{dx}$. Separate variables: $v\,dv = f(x)\,dx$ and integrate.
3. If $a = f(t)$: use $a = \frac{dv}{dt}$. Separate variables: $dv = f(t)\,dt$ and integrate.
4. Apply initial conditions to determine constants of integration.
5. If needed, find $x(t)$ by integrating $v = dx/dt$.

WORKED EXAMPLE 13 **Speed of the chain when it leaves the table**

A uniform chain of length $L = 2\,\text{m}$ and mass $M = 0.5\,\text{kg}$ is placed on a smooth horizontal table with $x_0 = 0.2\,\text{m}$ hanging over the edge. The chain is released from rest. Find the speed of the chain at the instant the last link leaves the table.

SOLUTION

Solution. Linear density $\lambda = M/L = 0.25\,\text{kg/m}$. Equation of motion for the whole chain: $a = \frac{g}{L}x = \frac{9.8}{2}x = 4.9x$. Since $a = f(x)$, use $a = v\frac{dv}{dx}$:

$$v\frac{dv}{dx} = 4.9x \quad \Rightarrow \quad v\,dv = 4.9x\,dx.$$

Integrate from initial state $(x = 0.2, v = 0)$ to final state $(x = L = 2, v = v_f)$:

$$\int_0^{v_f} v\,dv = 4.9\int_{0.2}^{2} x\,dx \;\Rightarrow\; \frac{1}{2}v_f^2 = 4.9\left[\frac{x^2}{2}\right]_{0.2}^{2} = 2.45\,(4 - 0.04) = 2.45 \times 3.96 = 9.702.$$

Thus $v_f^2 = 19.404 \Rightarrow v_f = \sqrt{19.404} \approx 4.40\,\text{m s}^{-1}$. Alternative energy method: The work done by gravity on the hanging part equals the gain in kinetic energy of the whole chain. The weight of the hanging part is $\lambda g x$, and it acts through a distance dx as x increases. Total work $= \int_{x_0}^{L} \lambda g x\,dx = \frac{1}{2}\lambda g(L^2 - x_0^2)$. Equating to $\frac{1}{2}Mv^2 = \frac{1}{2}\lambda L v^2$ gives $v = \sqrt{\frac{g}{L}(L^2 - x_0^2)} = \sqrt{gL - \frac{gx_0^2}{L}}$, which matches the integration result.

5.4 Variable Acceleration with Time-Dependent Force

Another common case is when the net force varies explicitly with time.

WORKED EXAMPLE 14 **Particle pulled by a time-varying force**

A particle of mass 2 kg rests on a smooth horizontal table. It is pulled by a horizontal force $F(t) = 6t$ N (where t is in seconds) applied via a light inextensible string. The particle starts from rest at $t = 0$. Find its velocity and displacement at $t = 3$ s.

SOLUTION

Solution. Newton's second law: $F = ma \Rightarrow 6t = 2a \Rightarrow a = 3t$. Since $a = f(t)$, use $a = \frac{dv}{dt}$:

$$\frac{dv}{dt} = 3t \quad \Rightarrow \quad dv = 3t\,dt.$$

Integrate from $t = 0$ ($v = 0$) to t:

$$\int_0^v dv = 3\int_0^t t\,dt \Rightarrow v = \frac{3}{2}t^2.$$

At $t = 3$ s: $v = \frac{3}{2} \times 9 = 13.5\,\text{m s}^{-1}$. Now find displacement: $v = \frac{dx}{dt} = \frac{3}{2}t^2 \Rightarrow dx = \frac{3}{2}t^2\,dt$. Integrate from $t = 0$ ($x = 0$) to $t = 3$:

$$x = \frac{3}{2}\int_0^3 t^2\,dt = \frac{3}{2}\left[\frac{t^3}{3}\right]_0^3 = \frac{1}{2} \times 27 = 13.5\,\text{m}.$$

5.5 Further Examples of Variable Acceleration

The chain problem and the time-varying force are prototypes. Other situations include:

- A rope being pulled up with a force that varies with the length already lifted.
- A particle attached to a spring on a rough surface where friction depends on position (though spring force is usually treated in the oscillations chapter).
- Systems where mass is added or removed (e.g., sand falling onto a moving conveyor belt) — treated via momentum, not $F = ma$ directly.

In all cases, the strategy is: **write Newton's second law for the system at a generic instant**, identify the variable acceleration, and integrate appropriately.

WARNING **Common mistake: using suvat for variable acceleration**

Never use $v^2 = u^2 + 2as$, $s = ut + \frac{1}{2}at^2$, etc., when acceleration is not constant. These equations are derived assuming constant a. If a varies with x or t, you must integrate using $a = v\,dv/dx$ or $a = dv/dt$.

WARNING **Common mistake: assuming a single tension throughout**

In a system with multiple string segments (separated by particles or pulleys), the tensions are generally different. Always label them T_1, T_2, … and write separate equations for each particle. Only if the string is continuous over a smooth pulley with no mass attached at the junction does the tension remain the same on both sides.

DID YOU KNOW? **Real-life application: Cable deployment**

The chain sliding off a table models the deployment of a cable from a drum, the unwinding of a hose, or the motion of a flexible rope in a pulley system. Engineers use the variable acceleration analysis to design braking systems that prevent excessive speeds when the driving weight increases. In Cameroon, similar principles apply to the design of cable-car systems in mountainous regions like the Mandara Mountains.

KEY POINTS **Key points for Section 5**

- For systems with ≥ 3 particles: write one $F = ma$ equation per particle; use the constraint of equal acceleration magnitude along each taut string segment.
- Different string segments → different tensions (T_1, T_2, …).
- Whole-system shortcut: $a = \frac{\sum \text{external driving forces}}{\sum \text{masses}}$ (valid when all parts share the same $|a|$).
- Movable pulleys introduce geometric constraints (e.g., $2a_{\text{pulley}} = a_1 + a_2$); treat only simple cases.
- Variable acceleration: if $a = f(x)$, use $v\,dv = f(x)\,dx$; if $a = f(t)$, use $dv = f(t)\,dt$. Integrate with initial conditions.
- Do not use suvat when a is not constant.

EXERCISE 5 **Practice problems**

1. Three particles of masses 2 kg, 3 kg and 5 kg are connected by light inextensible strings. The 2 kg and 3 kg masses lie on a smooth horizontal table, the 5 kg mass hangs over a smooth pulley at the edge. Find the acceleration and the two tensions.

2. A uniform chain of length 3 m and mass 1.2 kg rests on a smooth table with 0.5 m hanging over the edge. It is released from rest. Find the speed when the chain leaves the table.

3. A light movable pulley carries a mass of 6 kg. A light string passes over it; one end is fixed, the other is pulled upward with a constant force of 80 N. Find the acceleration of the 6 kg mass.

4. A particle of mass m on a smooth horizontal table is connected by a string passing over a pulley at the edge to a particle of mass $2m$ hanging vertically. A second string connects the hanging particle to a third particle of mass $3m$ hanging below it. Find the acceleration of the system and the tensions in both strings.

5. A particle of mass 3 kg on a smooth horizontal plane is acted upon by a force $F = (4t + 2)$ N in a fixed horizontal direction. If the particle starts from rest, find its velocity and displacement after 4 s.

SOLUTION **Exercise 1**

$m_1 = 2$, $m_2 = 3$ on table, $m_3 = 5$ hanging. Whole system: $a = \frac{5g}{2+3+5} = \frac{g}{2} = 4.9\ \text{m/s}^2$. Tensions: For m_1: $T_1 = 2a = 9.8$ N. For m_3: $5g - T_2 = 5a \Rightarrow T_2 = 5(g - a) = 24.5$ N.

SOLUTION **Exercise 2**

$\lambda = 0.4\ \text{kg/m}$, $L = 3$, $x_0 = 0.5$. $a = \frac{g}{L}x = \frac{9.8}{3}x$. $v\,dv = \frac{9.8}{3}x\,dx$. Integrate: $\frac{1}{2}v^2 = \frac{9.8}{6}(x^2 - 0.5^2)$. At $x = 3$: $v^2 = \frac{9.8}{3}(9 - 0.25) = \frac{9.8}{3} \times 8.75 = 28.583\cdots \Rightarrow v \approx 5.35\ \text{m/s}$.

SOLUTION **Exercise 3**

Constraint: $a_{\text{mass}} = a$, $a_{\text{free end}} = 2a$. Tension $T = F = 80$ N (massless free end). For mass 6 kg: $2T - 6g = 6a \Rightarrow 160 - 58.8 = 6a \Rightarrow a = 16.87\ \text{m/s}^2$ upward.

SOLUTION **Exercise 4**

Masses: m (table), $2m$ (upper hanging), $3m$ (lower hanging). Strings: T_1 between m and $2m$, T_2 between $2m$ and $3m$. Equations: m: $T_1 = ma$. $2m$: $T_2 + 2mg - T_1 = 2ma$. $3m$: $3mg - T_2 = 3ma$. Add: $5mg = 6ma \Rightarrow a = \frac{5g}{6}$. $T_1 = m\frac{5g}{6} = \frac{5mg}{6}$. $T_2 = 3mg - 3ma = 3mg - 3m\frac{5g}{6} = \frac{mg}{2}$.

SOLUTION **Exercise 5**

$m = 3\,\text{kg}$, $F = 4t + 2$. $a = \frac{F}{m} = \frac{4t+2}{3}$. $dv = \frac{4t+2}{3}dt$. Integrate 0 to t: $v = \frac{1}{3}(2t^2 + 2t) = \frac{2}{3}t(t+1)$. At $t = 4$: $v = \frac{2}{3} \times 4 \times 5 = \frac{40}{3} \approx 13.3\,\text{m/s}$. $x = \int_0^4 v\,dt = \frac{2}{3}\int_0^4 (t^2 + t)\,dt = \frac{2}{3}\left[\frac{t^3}{3} + \frac{t^2}{2}\right]_0^4 = \frac{2}{3}\left(\frac{64}{3} + 8\right) = \frac{2}{3} \times \frac{88}{3} = \frac{176}{9} \approx 19.6\,\text{m}$.

6 SECTION 6: ENERGY METHODS AND REAL-LIFE PROBLEMS

In the previous section we analysed pulley systems with more than two masses and even cases where the acceleration varies with time. We solved those problems by writing Newton's second law for each particle, eliminating the tensions, and integrating when necessary. That approach is powerful because it gives the acceleration and the tension at every instant. However, when the question only asks for the speed after a given displacement (or the distance needed to reach a certain speed), there is a much shorter route: the **work–energy principle**. In this section we develop energy methods for connected particles, show how they bypass the internal tensions entirely, and apply them to realistic engineering situations such as lifts, cable cars and towed vehicles.

6.1 Work, Kinetic Energy and Potential Energy in Connected Systems

DEFINITION **Work–Energy Principle for a System of Particles**

For any system of particles, the change in the total kinetic energy equals the work done by all forces (external and internal) acting on the system:

$$\Delta K_{\text{total}} = W_{\text{external}} + W_{\text{internal}}.$$

If the internal forces are conservative (e.g. gravity within the system) we can move their work to the left-hand side as a change in potential energy. For a system subject to gravity and possibly friction, the principle becomes

$$\Delta K + \Delta U_{\text{grav}} = W_{\text{friction}} + W_{\text{other external}}.$$

In a connected system with a **light inextensible string** passing over a **smooth pulley**, the tensions are internal forces. Because the string is inextensible, the displacement of each particle along the string is the same in magnitude. The tension on one particle does positive work while the tension on the other does an equal amount of negative work. Hence:

PROPERTY **Tensions Do No Net Work in a Light Inextensible String**

In a connected system where the string is light and inextensible and the pulley is smooth, the total work done by the tension forces is zero. Tensions merely transfer energy between the particles without dissipating or creating it.

This property is the key to the energy method: we can write a single energy equation for the whole system without ever calculating the tension.

6.2 Conservation of Mechanical Energy (Smooth Contacts)

When all contacts are smooth (no friction) and the pulley is smooth, the only external forces doing work are gravity (conservative) and possibly a driving force. If no non-conservative external forces act, mechanical energy is conserved.

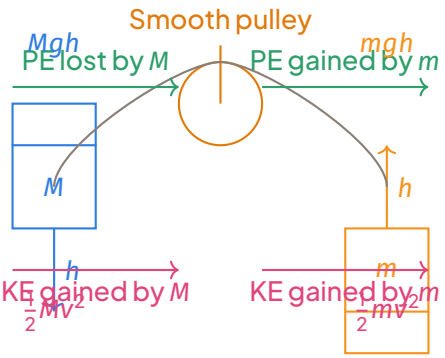
KEY POINTS **Energy Conservation for a Two-Mass Atwood-Type System**

Consider two masses M and m ($M > m$) connected by a light inextensible string over a smooth pulley. If the system is released from rest and the heavier mass descends a distance h, then:

- Loss of gravitational PE of M: Mgh.
- Gain of gravitational PE of m: mgh.
- Gain of kinetic energy of both masses: $\frac{1}{2}(M+m)v^2$.

Conservation of mechanical energy gives:

$$Mgh = mgh + \frac{1}{2}(M+m)v^2 \quad \Longrightarrow \quad (M-m)gh = \frac{1}{2}(M+m)v^2.$$



Energy-flow diagram for an Atwood's machine: the potential energy lost by the descending mass M is converted into kinetic energy of both masses and potential energy gained by the rising mass m.

WARNING **Common Mistake: Forgetting the Kinetic Energy of the Rising Mass**

In a connected system every moving particle carries kinetic energy. A frequent error is to write $Mgh = \frac{1}{2}Mv^2$, ignoring the fact that the lighter mass m also moves and therefore has kinetic energy $\frac{1}{2}mv^2$. Always include the KE of all particles that are in motion.

6.3 Work–Energy Principle with Friction (Rough Surfaces)

When one or more particles move on rough surfaces, friction does negative work. Mechanical energy is no longer conserved; we must use the full work–energy principle.

PROPERTY **Work–Energy Principle with Friction**

For a connected system where a particle of mass m moves a distance s on a rough horizontal surface (coefficient of friction μ), the work done against friction is $W_{\text{fric}} = -\mu mgs$. The energy equation becomes:

(Loss of PE of descending masses) = (Gain of KE of all masses)+(Gain of PE of rising masses)+μmgs.

Equivalently:

$$\Delta K + \Delta U = W_{\text{friction}} \quad \text{(where } W_{\text{friction}} < 0\text{)}.$$

METHOD **Energy Route for Connected Particles**

1. Identify the initial and final states. Choose a reference level for gravitational potential energy (usually the lowest point reached by any particle).

2. Compute the total change in gravitational potential energy ΔU for all particles. $\Delta U = \sum m_i g \Delta h_i$ (positive if the particle rises).
3. Compute the total change in kinetic energy ΔK for all particles. If starting from rest, $\Delta K = \frac{1}{2}\sum m_i v^2$.
4. Account for work done against friction. For each particle on a rough surface, $W_{\text{fric}} = -\mu R \times$ distance moved.
5. Write the work–energy equation: $\Delta K = -\Delta U + W_{\text{fric}} + W_{\text{other external}}$.
6. Solve for the unknown (usually v or the distance s).

WORKED EXAMPLE 15 **Block on Rough Table Connected to Hanging Mass**

A block of mass 4 kg rests on a rough horizontal table ($\mu = 0.2$). It is connected by a light inextensible string passing over a smooth pulley at the edge of the table to a hanging mass of 2 kg. The system is released from rest. Find the speed of the masses after the hanging mass has descended 1.5 m.

SOLUTION

Solution by Energy Method.

- Initial state: system at rest, hanging mass at reference level (PE = 0). Block on table: PE constant (horizontal motion).
- Final state: hanging mass descended $h = 1.5$ m (PE lost = $mgh = 2 \times 9.8 \times 1.5 = 29.4$ J). Block moved 1.5 m on table.
- Friction on block: $R = 4g = 39.2$ N, $F_{\text{fric}} = \mu R = 0.2 \times 39.2 = 7.84$ N. Work done against friction $W_{\text{fric}} = -7.84 \times 1.5 = -11.76$ J.
- Both masses move with same speed v. Total KE gain = $\frac{1}{2}(4+2)v^2 = 3v^2$.
- Energy equation: $\Delta K = -\Delta U + W_{\text{fric}}$.

$$3v^2 = 29.4 - 11.76 = 17.64 \quad \Rightarrow \quad v^2 = 5.88 \quad \Rightarrow \quad v = \sqrt{5.88} \approx 2.42 \text{ m s}^{-1}.$$

Check with Newton's Laws (optional). For hanging mass: $2g - T = 2a$. For block: $T - \mu(4g) = 4a$. Adding: $2g - 0.2(4g) = 6a \Rightarrow a = \frac{19.6 - 7.84}{6} = 1.96 \text{ m s}^{-2}$. Then $v^2 = 2as = 2 \times 1.96 \times 1.5 = 5.88$, same result. The energy method avoided finding T entirely.

6.4 Comparing Newton's Laws and Energy Methods

Aspect	Newton's Second Law ($F = ma$)	Work–Energy Principle
Unknowns	Gives acceleration a and tension T at every instant.	Gives speed v after a given displacement s (or s for a given v).
Forces needed	Must identify all forces on each particle, including internal tensions.	Only external forces that do work (gravity, friction, driving forces). Tensions cancel out.
Variable acceleration	Handles naturally via $a = v\frac{dv}{dx}$ or integration.	Handles naturally if work integrals can be evaluated.
When to prefer	• Need tension or acceleration at a specific instant. • Forces change direction (e.g. particle leaves surface). • Motion on curved paths where normal reaction does no work but affects constraints.	• Only speed after a given displacement is asked. • System has many particles; writing many $F = ma$ equations is tedious. • Friction is present but constant (work = force × distance).

KEY POINTS **Rule of Thumb**

- "Find the acceleration / tension" → Newton's laws.
- "Find the speed after moving x metres" → Energy method (usually faster).
- "Find the distance to come to rest" → Energy method.

6.5 Power in Connected Systems (GCE Extension)

Power is the rate at which work is done. For a constant force F acting on a particle moving with velocity v in the direction of the force, $P = Fv$. In a connected system, the power delivered by a driving force (e.g. a motor pulling a cable) equals the rate of increase of the total mechanical energy of the system plus the rate of working against friction.

EXEMPLE **Power of a Winch**

A winch pulls a loaded cage of mass 500 kg up a rough inclined plane at constant speed $2\,\mathrm{m\,s^{-1}}$. The plane is at 30° to the horizontal and $\mu = 0.15$. Find the power output of the winch.

At constant speed, $\Delta K = 0$. In 1 s the cage moves 2 m up the plane.

- Gain in PE: $mgh = 500 \times 9.8 \times (2\sin 30^\circ) = 4900$ J.

- Work against friction: $F_{\text{fric}} = \mu mg\cos 30^\circ = 0.15 \times 500 \times 9.8 \times \frac{\sqrt{3}}{2} \approx 636.5\,\text{N}$. Work in 1 s: $636.5 \times 2 = 1273\,\text{J}$.
- Total work per second (power) $P = 4900 + 1273 = 6173\,\text{W} \approx 6.17\,\text{kW}$.

Alternatively, tension in cable $T = mg\sin 30^\circ + \mu mg\cos 30^\circ \approx 3086.5\,\text{N}$, then $P = Tv = 3086.5 \times 2 = 6173\,\text{W}$.

6.6 Real-Life Modelling Problems

The GCE examination often presents a practical situation (a lift, a cable car, a towed vehicle) and expects you to go through the full **modelling cycle**.

METHOD **Modelling Cycle for Real-Life Problems**

1. State assumptions explicitly. Typical assumptions: light inextensible strings/cables, smooth pulleys, particles (no rotation), constant $g = 9.8\,\text{m s}^{-2}$, no air resistance, constant coefficient of friction.
2. Draw a clear diagram showing all forces, displacements, and coordinate directions.
3. Formulate equations using either Newton's laws or energy principles (choose the most efficient).
4. Solve mathematically for the required quantities.
5. Interpret the answer in the context of the problem (units, direction, physical plausibility).
6. Critique the model: How would the answer change if we relaxed an assumption? (e.g. massive cable → tension varies along cable; pulley friction → tensions on either side differ; air resistance → terminal velocity).

WORKED EXAMPLE 16 **Lift with Counterweight — A Classic Application**

A lift of mass 1200 kg is connected by a light inextensible cable passing over a smooth pulley to a counterweight of mass 950 kg. The lift carries passengers of total mass 350 kg. The system starts from rest. Find:

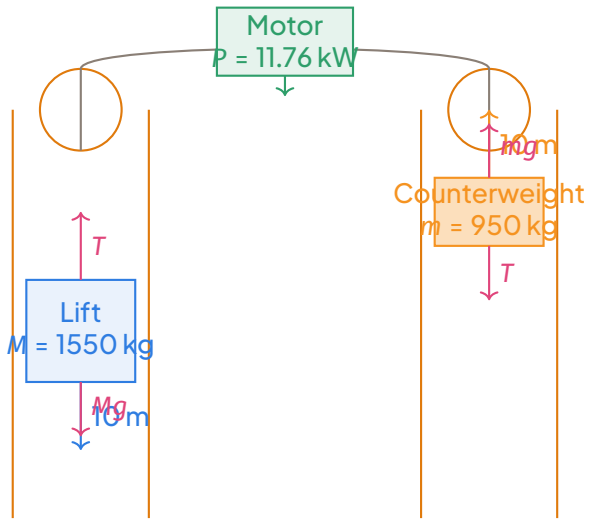
1. The acceleration of the lift.
2. The tension in the cable.
3. The speed of the lift after it has moved 10 m.

4. The power required from the motor to maintain a constant upward speed of $2\,\text{m s}^{-1}$ when the lift is fully loaded (assume the motor drives the pulley).

SOLUTION

Assumptions: Cable light and inextensible, pulley smooth, lift and counterweight move vertically as particles, $g = 9.8\,\text{m s}^{-2}$.
Diagram: Lift (total mass $M = 1200 + 350 = 1550$ kg) on left, counterweight $m = 950$ kg on right. Lift moves up, counterweight moves down.
(i) Acceleration (Newton's Laws). Let upward be positive for both masses. Lift: $T - Mg = Ma$. Counterweight: $T - mg = m(-a) \Rightarrow mg - T = ma$. Add: $(m - M)g = (M + m)a \Rightarrow a = \frac{(950-1550)\times 9.8}{2500} = -2.352\,\text{m s}^{-2}$. The negative sign means the lift accelerates downwards (the counterweight is lighter than the loaded lift). Magnitude $a = 2.35\,\text{m s}^{-2}$ downwards.
(ii) Tension. $T = M(g + a) = 1550(9.8 - 2.352) = 1550 \times 7.448 = 11544.4$ N ≈ 11.5 kN. (Check with counterweight: $T = m(g - a) = 950(9.8 + 2.352) = 950 \times 12.152 = 11544.4$ N. Consistent.)
(iii) Speed after 10 m (Energy Method). Lift moves down 10 m, counterweight moves up 10 m. Loss of PE of lift: $Mgh = 1550 \times 9.8 \times 10 = 151900$ J. Gain of PE of counterweight: $mgh = 950 \times 9.8 \times 10 = 93100$ J. Net loss of PE = $151900 - 93100 = 58800$ J. This becomes KE of both: $\frac{1}{2}(M + m)v^2 = \frac{1}{2}(2500)v^2 = 1250v^2$. $1250v^2 = 58800 \Rightarrow v^2 = 47.04 \Rightarrow v = 6.86\,\text{m s}^{-1}$.
(iv) Power for Constant Upward Speed $2\,\text{m s}^{-1}$. At constant speed, $a = 0$. The motor must overcome the weight difference. Net force the motor must overcome: $Mg - mg = (1550 - 950) \times 9.8 = 5880$ N. Power $P = Fv = 5880 \times 2 = 11760$ W $= 11.76$ kW. (Alternatively: rate of gain of PE of lift = $Mgv = 15190 \times 2 = 30380$ W; rate of loss of PE of counterweight = $mgv = 9310 \times 2 = 18620$ W; net power needed = $30380 - 18620 = 11760$ W.)
Critique of Model:

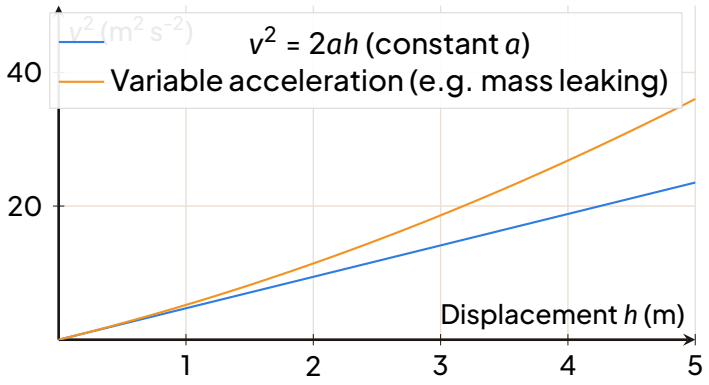
- If the cable has mass, tension varies along its length; the motor would need extra power to lift the cable itself.
- If the pulley has friction, the tensions on either side differ even without a motor; the motor must overcome both weight difference and pulley friction.
- Air resistance on the lift would increase the required power at high speeds.



Lift with counterweight: forces, displacements and motor power. The motor supplies the power to overcome the weight difference when moving at constant speed.

6.7 Variable Acceleration and Energy: v^2 vs Displacement Graph

For a system with constant acceleration (like the Atwood machine with constant masses), the energy equation $(M - m)gh = \frac{1}{2}(M + m)v^2$ gives $v^2 = \frac{2(M-m)g}{M+m}h$. This is a straight line through the origin when v^2 is plotted against h.



Graph of v^2 against displacement h. For constant acceleration (smooth Atwood machine) the graph is a straight line through the origin with slope $2a$. Curvature indicates variable acceleration (e.g. mass changing, friction varying).

WARNING **Common Mistake: Using Conservation of Energy When Friction Acts**

If any non-conservative force (friction, air resistance, a motor doing work) acts on the

system, mechanical energy is not conserved. You must use the work–energy principle:

$$\Delta K + \Delta U = W_{\text{non-conservative}}.$$

Writing "Loss of PE = Gain of KE" when friction is present will give an answer that is too large (it predicts a higher speed than reality).

WORKED EXAMPLE 17 **Towed Vehicle on a Hill**

A car of mass 1200 kg tows a trailer of mass 800 kg up a hill inclined at $\sin^{-1}(0.1)$ to the horizontal. The resistance to motion (constant) is 400 N on the car and 250 N on the trailer. The driving force of the car's engine is 5000 N. The tow-bar is light and inextensible. Find:

1. The acceleration of the system.
2. The tension in the tow-bar.
3. The distance travelled from rest to reach a speed of $15\ \text{m s}^{-1}$.

SOLUTION

Assumptions: Car and trailer as particles, tow-bar light and inextensible, constant resistances, $g = 9.8$.
Diagram: Car (front) pulling trailer. Incline angle θ, $\sin\theta = 0.1$, $\cos\theta = \sqrt{0.99} \approx 0.995$.
(i) Acceleration (Newton's Laws — needed for tension). System as a whole (mass 2000 kg): Driving force − total resistance − component of weight down plane = Ma. $5000 - (400 + 250) - 2000g\sin\theta = 2000a$. $5000 - 650 - 2000 \times 9.8 \times 0.1 = 2000a$. $5000 - 650 - 1960 = 2000a \Rightarrow 2390 = 2000a \Rightarrow a = 1.195\ \text{m s}^{-2}$.
(ii) Tension in tow-bar. Consider trailer only (mass 800 kg): $T - 250 - 800g\sin\theta = 800a$. $T = 250 + 800 \times 9.8 \times 0.1 + 800 \times 1.195 = 250 + 784 + 956 = 1990$ N.
(iii) Distance to reach $15\ \text{m s}^{-1}$ (Energy Method — faster). Work done by engine = $5000 \times s$. Work against resistances = $-(400 + 250)s = -650s$. Gain in PE = $(1200 + 800)g(s\sin\theta) = 2000 \times 9.8 \times 0.1s = 1960s$. Gain in KE = $\frac{1}{2}(2000)(15^2) = 225000$ J. Work–energy: $\Delta K = W_{\text{engine}} + W_{\text{resist}} + W_{\text{weight}}$. $225000 = 5000s - 650s - 1960s = 2390s$. $s = \frac{225000}{2390} \approx 94.1$ m.
(Check: $v^2 = 2as = 2 \times 1.195 \times 94.1 \approx 225$, correct.)

6.8 Further Real-Life Examples

EXEMPLE **Cable Car**

A cable car of mass 2000 kg carries 20 passengers (average mass 70 kg). It is pulled up a slope of 30° by a cable driven by a motor at the top. The cable is light and inextensible.

The resistance to motion is 1500 N. Find the power required to maintain a constant speed of 6 m s^{-1}.

Total mass $M = 2000+20\times70 = 3400$ kg. At constant speed, tension $T = Mg\sin 30^\circ + 1500 = 3400 \times 9.8 \times 0.5 + 1500 = 16660 + 1500 = 18160$ N. Power $P = Tv = 18160 \times 6 = 108960$ W ≈ 109 kW.

EXEMPLE **Bucket Raised from a Well — Variable Mass**

A bucket of mass 5 kg (including water) is raised vertically from a well 20 m deep by a rope wound on a winch. The rope is light and inextensible. Water leaks out at a constant rate so that the bucket is empty at the top. The mass of the empty bucket is 2 kg. Find the work done by the winch if the bucket is raised at constant speed.

Mass varies linearly from 5 kg to 2 kg over 20 m. Average mass $= \frac{5+2}{2} = 3.5$ kg. Work done against gravity = (average weight) × distance $= 3.5g \times 20 = 3.5 \times 9.8 \times 20 = 686$ J. (Alternatively, integrate: $m(x) = 5 - \frac{3}{20}x$, $W = \int_0^{20} m(x)g\,dx = g[5x - \frac{3}{40}x^2]_0^{20} = 9.8(100 - 30) = 686$ J.) This is an example of **variable acceleration** (if speed were not constant) handled by energy integration.

EXEMPLE **Load Sliding Down a Rough Chute**

A package of mass 10 kg slides down a rough chute inclined at 25° to the horizontal. The coefficient of friction is 0.3. The package starts from rest at the top and reaches the bottom with speed 8 m s^{-1}. Find the length of the chute.

Energy method: Loss of PE = Gain in KE + Work against friction. $mgh = \frac{1}{2}mv^2 + \mu mg\cos\theta \times s$, where $h = s\sin\theta$. $mgs\sin\theta = \frac{1}{2}mv^2 + \mu mg\cos\theta\, s$. Cancel m: $gs\sin\theta = \frac{1}{2}v^2 + \mu g\cos\theta\, s$. $s(g\sin\theta - \mu g\cos\theta) = \frac{1}{2}v^2$. $s = \frac{v^2}{2g(\sin\theta - \mu\cos\theta)} = \frac{64}{2\times9.8(\sin 25^\circ - 0.3\cos 25^\circ)}$. $\sin 25^\circ \approx 0.423$, $\cos 25^\circ \approx 0.906$. Denominator $= 19.6(0.423 - 0.272) = 19.6 \times 0.151 = 2.96$. $s \approx 21.6$ m.

6.9 Summary and Examination Tips

KEY POINTS **Key Points for the GCE Examination**

- Energy conservation applies only when all contacts are smooth and no external non-conservative forces do work. Equation: $\Delta K + \Delta U = 0$.

- Work–energy principle is the general tool: $\Delta K + \Delta U = W_{\text{friction}} + W_{\text{driving}}$.
- Tensions in a light inextensible string do no net work — they are internal forces that cancel out in the system energy equation.
- Always include the kinetic energy of every moving particle.
- Choose the method wisely: Newton's laws for acceleration/tension; energy for speed/distance.
- Modelling cycle: Assumptions → Diagram → Equations → Solution → Interpretation → Critique.
- Power $P = Fv$ for constant force and velocity; for variable force, $P = \frac{dW}{dt}$.

EXERCISE 6 **Practice Problems**

- Two masses 3 kg and 5 kg are connected by a light inextensible string over a smooth pulley. The 5 kg mass is held at rest on a rough horizontal table ($\mu = 0.4$) and the 3 kg mass hangs freely. The system is released. Use energy methods to find the speed of the masses after the hanging mass has fallen 1.2 m.
- A train of mass 400 tonnes climbs a slope of 1 in 100 at a constant speed of 20 m s^{-1}. The resistance to motion is 50 N per tonne. Find the power of the engine in kW.
- A skier of mass 75 kg starts from rest at the top of a rough slope of length 200 m inclined at 15°. The coefficient of friction is 0.08. Find the skier's speed at the bottom. If the skier then travels on horizontal snow with the same coefficient of friction, how far does she travel before stopping?
- A lift of mass 800 kg is counterbalanced by a mass of 600 kg. The lift carries 10 people of average mass 70 kg. The system starts from rest. (a) Find the acceleration. (b) Find the tension in the cable. (c) Use energy to find the speed after the lift has descended 15 m. (d) What power must the motor provide to raise the loaded lift at a constant 2 m s^{-1}?

SOLUTION **Exercise 1**

Loss of PE of 3 kg mass $= 3g(1.2) = 35.28$ J. Work against friction on 5 kg mass $= \mu mgs = 0.4 \times 5 \times 9.8 \times 1.2 = 23.52$ J. Gain in KE of both masses $= \frac{1}{2}(3+5)v^2 = 4v^2$. Energy: $35.28 = 4v^2 + 23.52 \Rightarrow 4v^2 = 11.76 \Rightarrow v^2 = 2.94 \Rightarrow v = 1.71$ m s^{-1}.

SOLUTION **Exercise 2**

Mass = 400 000 kg. Slope $\sin\theta = 1/100 = 0.01$. Component of weight down slope $= Mg\sin\theta = 400\,000 \times 9.8 \times 0.01 = 39\,200$ N. Resistance = $50 \times 400 = 20\,000$ N. Total force to overcome at constant speed = 39 200 + 20 000 = 59 200 N. Power $P = Fv = 59\,200 \times 20 = 1\,184\,000$ W = 1184 kW.

SOLUTION **Exercise 3**

Slope: loss of PE = $mgh = 75 \times 9.8 \times (200\sin 15^\circ) = 75 \times 9.8 \times 51.76 = 38\,041$ J. Work against friction on slope = $\mu mg\cos 15^\circ \times 200 = 0.08 \times 75 \times 9.8 \times 0.966 \times 200 = 11\,370$ J. KE at bottom = 38 041 − 11 370 = 26 671 J. $v = \sqrt{2 \times 26\,671/75} = 26.7\ \text{m s}^{-1}$. Horizontal: initial KE = 26 671 J. Work against friction = $\mu mgd = 0.08 \times 75 \times 9.8 \times d = 58.8d$. $58.8d = 26\,671 \Rightarrow d = 453$ m.

SOLUTION **Exercise 4**

(a) Total lift mass $M = 800 + 700 = 1500$ kg. Counterweight $m = 600$ kg. $a = \frac{(M-m)g}{M+m} = \frac{900\times 9.8}{2100} = 4.2\ \text{m s}^{-2}$ (lift down). (b) $T = M(g-a) = 1500(9.8-4.2) = 8400$ N. (c) Lift descends 15 m, counterweight rises 15 m. Net PE loss = $(M-m)gh = 900 \times 9.8 \times 15 = 132\,300$ J. KE gain = $\frac{1}{2}(2100)v^2 = 1050v^2$. $1050v^2 = 132\,300 \Rightarrow v^2 = 126 \Rightarrow v = 11.2\ \text{m s}^{-1}$. (d) Constant speed up: motor must overcome weight difference = $(M-m)g = 8820$ N. $P = Fv = 8820 \times 2 = 17\,640$ W = 17.6 kW.

DID YOU KNOW? **Historical Note: The Atwood Machine**

The Atwood machine was invented in 1784 by the English mathematician George Atwood (1745–1807) as a laboratory device to verify Newton's laws of motion with reduced acceleration (so that motion could be timed accurately with the clocks of the era). By choosing two nearly equal masses, the acceleration $a = \frac{M-m}{M+m}g$ can be made very small, allowing precise measurement of g. Modern physics laboratories still use a version of Atwood's machine, often with a photogate timer, to teach dynamics and energy conservation.

DID YOU KNOW? **Cameroon Context: The Edea–Douala Cable Car Project**

Feasibility studies have been conducted for a cable car system linking Douala and Edea to ease traffic congestion on the national highway. Such a system would involve cabins of mass ~ 2000 kg carrying up to 30 passengers, moving on cables with a driving motor at one station. The power calculations we have just performed are exactly what mechanical

engineers carry out to size the motors and cables, taking into account the weight difference between ascending and descending cabins, friction in the pulleys, wind loads, and safety factors. The “modelling cycle” you learn here is the same cycle used in real engineering design offices in Yaoundé and Douala.

7 INTEGRATION ACTIVITY

7.1 Aims of the activity

This integration activity closes the chapter by bringing together, in one realistic modelling task, every skill developed in Sections 1 to 6:

- stating and justifying **modelling assumptions** (light inextensible rope, smooth pulley, particle model);
- drawing correct **free-body diagrams** for each body of a connected system;
- applying **Newton's second law** to each particle and solving the resulting simultaneous equations for the **acceleration** and the **tension**;
- using **suvat equations** for constant-acceleration stages;
- analysing **multi-stage motion** (what happens when one body suddenly stops or the rope goes slack);
- applying the **work–energy principle** as an independent check of kinematic results;
- resolving forces on a **rough inclined plane** and deciding whether motion is possible;
- **critiquing a model**: discussing the effect of rope mass, pulley friction and impulsive jerks.

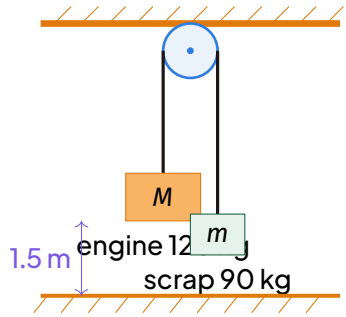
By the end, each group will have carried out a complete modelling cycle: real situation → assumptions → equations → solution → interpretation → critique.

7.2 Situation

The Limbe Engine Hoist.

Mr. Ekane runs a small vehicle-repair workshop in Limbe, at the foot of Mount Cameroon. He must remove a car engine of mass $M = 120$ kg and lower it gently to the workshop floor. Having no crane, he passes a strong rope over a smooth horizontal steel beam fixed 4.0 m above the floor. To one end of the rope he attaches the engine; to the other end he attaches a **counterweight** made of scrap metal of mass $m = 90$ kg. He releases the system from rest, with the engine initially 1.5 m above the floor.

The rope is rated by its manufacturer for a maximum safe tension of 1500 N. Take $g = 9.8\ \text{m s}^{-2}$ throughout.



The engine hoist: engine M and scrap counterweight m connected by a rope passing over a smooth beam.

In a **second stage** of the job, Mr. Ekane wonders whether the same counterweight could later drag the engine back up a rough concrete ramp inclined at 20° to the horizontal, for which the coefficient of friction between engine and ramp is $\mu = 0.3$ (the rope still passes over the smooth beam, with the counterweight hanging vertically).

Working in groups, you are the consulting team. Produce a full written report answering the tasks below, ending with a critical discussion of the model.

7.3 Tasks

1. **Modelling.** State at least four modelling assumptions needed to treat this as a standard connected-particles problem. Draw the free-body diagram of the engine and that of the counterweight, showing all forces.

2. **First stage of motion.** Taking the engine and the counterweight separately, apply Newton's second law and hence find:

   (a) the acceleration of the system;

   (b) the tension T in the rope.

   Is the rope safe? Justify with an inequality.

3. **Speed after** 1.5 m**.** The engine descends 1.5 m from rest before touching the floor.

   (a) Find its speed on reaching the floor using a suvat equation.

   (b) Find the same speed using the work–energy principle applied to the whole system (loss of gravitational potential energy = gain of kinetic energy). Compare the two answers.

4. **Second stage: rope goes slack.** Once the engine rests on the floor, the counterweight is still moving upwards. Explain why the rope becomes slack, state the counterweight's subsequent acceleration, and find the extra height it rises before momentary rest. Hence find its greatest height above its position at the moment of release.

5. **Extension: the rough ramp.** For the return job, the engine lies on a ramp inclined at 20° ($\mu = 0.3$) and is attached to the same 90 kg counterweight hanging vertically over the beam.

(a) Determine, with full working, whether the 90 kg counterweight can pull the engine up the ramp; if it can, find the acceleration of the system.

(b) Find the minimum counterweight mass $m_{\min}$ that would just set the engine moving up the ramp.

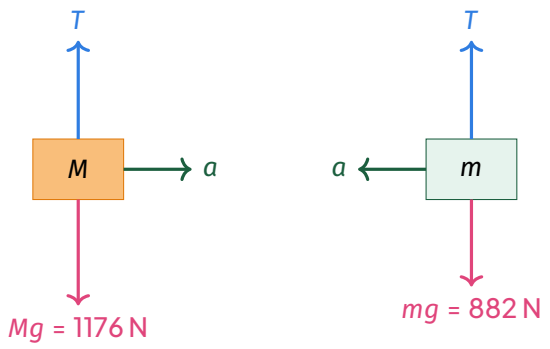
6. **Critique.** Discuss how each of the following would change your numerical answers: the rope has mass; the beam is rough (friction at the axle); the engine is snatched suddenly rather than released smoothly. Would the true safe working tension of the rope need to be higher or lower than your computed T?

7.4 Model answer

1. Modelling assumptions and free-body diagrams.

Assumptions: the rope is **light** (mass negligible) and **inextensible** (both bodies have the same magnitude of acceleration); the beam acts as a **smooth pulley** (same tension on both sides); the bodies are modelled as **particles**; air resistance is neglected; g is constant.

Forces on the engine: weight $Mg = 120 \times 9.8 = 1176$ N downwards, tension T upwards. Forces on the counterweight: weight $mg = 90 \times 9.8 = 882$ N downwards, tension T upwards.



Free-body diagrams of the engine (left) and the counterweight (right).

2. Acceleration and tension.

The engine (heavier) descends; take the direction of motion as positive for each body.

Engine (↓ positive): $Mg - T = Ma$, i.e.

$$1176 - T = 120a. \tag{1}$$

Counterweight (↑ positive): $T - mg = ma$, i.e.

$$T - 882 = 90a. \tag{2}$$

Adding (1) and (2) eliminates T:

$$1176 - 882 = (120 + 90)a \implies a = \frac{294}{210} = 1.4\ \text{m s}^{-2}.$$

Substituting in (2):

$$T = 882 + 90(1.4) = 882 + 126 = 1008\ \text{N}.$$

Safety check: $T = 1008$ N < 1500 N. **The rope is safe**, with a safety margin of 492 N (a load factor of $\frac{1500}{1008} \approx 1.49$).

3. Speed after descending 1.5 m.

(a) suvat. With $u = 0$, $a = 1.4\ \text{m s}^{-2}$, $s = 1.5$ m:

$$v^2 = u^2 + 2as = 0 + 2(1.4)(1.5) = 4.2 \implies v = \sqrt{4.2} \approx 2.05\ \text{m s}^{-1}.$$

(b) Energy method. For the whole system, the tension does no net work (it is an internal force and the rope is inextensible, so the works done by T on the two bodies cancel). The engine loses Mgh of potential energy while the counterweight gains mgh; both bodies gain kinetic energy, a total of $\frac{1}{2}(M + m)v^2$:

$$Mgh - mgh = \tfrac{1}{2}(M + m)v^2$$

$$(120 - 90)(9.8)(1.5) = \tfrac{1}{2}(210)v^2$$

$$441 = 105\,v^2 \implies v^2 = 4.2 \implies v \approx 2.05\ \text{m s}^{-1}.$$

The two methods agree exactly, as expected: the suvat result uses the acceleration obtained from Newton's second law, and for this conservative system Newton's second law and energy conservation are two statements of the same physics. Agreement of the two methods is a valuable **independent check** in the examination.

4. Second stage: the counterweight after the engine lands.

When the engine touches the floor, the floor exerts a normal reaction on it; the engine can no longer pull the rope down. The rope goes **slack** ($T = 0$), and the counterweight, moving upwards at $v = 2.05\ \text{m s}^{-1}$, is now in free motion under gravity alone: its acceleration is $-g = -9.8\ \text{m s}^{-2}$ (taking upwards as positive).

Extra height h' risen before momentary rest (final speed 0):

$$0 = v^2 - 2gh' \implies h' = \frac{4.2}{2 \times 9.8} = \frac{4.2}{19.6} \approx 0.214\ \text{m}.$$

During the first stage the counterweight rose 1.5 m, so its greatest height above its release position is

$$1.5 + 0.214 \approx 1.71\ \text{m}.$$

Consistency check: the counterweight started about 2.5 m below the beam (the beam is 4.0 m above the floor and the engine hung 1.5 m above the floor), so even after rising 1.71 m it does not strike the beam: the model remains consistent.

5. Extension: the rough 20° **ramp.**

Resolve the engine's weight along and perpendicular to the ramp:

$$Mg \sin 20^\circ = 1176 \times 0.342 \approx 402.2\ \text{N}, \qquad R = Mg \cos 20^\circ = 1176 \times 0.940 \approx 1105.0\ \text{N}.$$

Limiting friction (acting down the ramp, opposing the intended upward motion):

$$F_{\max} = \mu R = 0.3 \times 1105.0 \approx 331.5\ \text{N}.$$

For the engine to be on the point of moving up the ramp, the rope must pull it up with a tension at least equal to the sum of the downslope component of the weight and the limiting friction:

$$T_{\text{needed}} = Mg\sin 20^\circ + \mu Mg\cos 20^\circ = 402.2 + 331.5 = 733.7\ \text{N}.$$

(a) Can the 90 kg counterweight do the job? For the counterweight to pull the engine up, the counterweight must descend, and the tension it can transmit to the rope is then less than its weight $mg = 882$ N (since $mg - T = ma$ with $a > 0$). Motion is possible if and only if the driving weight exceeds the total resistance:

$$mg = 882\ \text{N} > 733.7\ \text{N} = T_{\text{needed}}.$$

The condition is satisfied, so the 90 kg counterweight can pull the engine up the ramp. Applying Newton's second law to the whole system in the direction of motion,

$$mg - Mg\sin 20^\circ - \mu Mg\cos 20^\circ = (M+m)a$$

$$a = \frac{882 - 733.7}{210} = \frac{148.3}{210} \approx 0.71\ \text{m s}^{-2}.$$

(The corresponding tension is $T = m(g-a) = 90(9.8 - 0.71) \approx 818$ N, comfortably below the rope rating.)

(b) Minimum counterweight mass. The engine is just on the point of moving up when the system is in limiting equilibrium, i.e. $m_{\min}g = Mg\sin 20^\circ + \mu Mg\cos 20^\circ$:

$$m_{\min} = M(\sin 20^\circ + \mu\cos 20^\circ) = 120(0.342 + 0.3\times 0.940) = 120(0.624) \approx 74.9\ \text{kg}.$$

So any counterweight heavier than about 75 kg suffices; the existing 90 kg of scrap is adequate, with margin.

WARNING **A trap in Task 5**

Many students assume that because the 90 kg counterweight is lighter than the 120 kg engine, it cannot drag the engine up the ramp. Always compute the actual driving and resisting forces: on a 20° ramp only the component $Mg\sin 20^\circ$ (plus friction) must be overcome, not the full weight Mg. Here the required force is only about 62% of the engine's weight.

6. Critique of the model.

- Rope mass. A real rope of several metres has non-negligible mass; part of the driving force is used to accelerate the rope itself, so the true acceleration is slightly smaller than $1.4\ \text{m s}^{-2}$ and the tension varies along the rope (greatest at the top, near the beam).
- Friction at the beam. A rough axle makes the tensions on the two sides unequal; part of the weight difference is "wasted" overcoming this friction, reducing a. The tension on the engine side could then exceed our computed 1008 N.
- Sudden jerks. If the rope is snatched, or the engine swings and is caught abruptly, impulsive tensions occur which can momentarily far exceed the steady value. For this reason real

safety codes require the rated tension of lifting equipment to be a generous multiple of the computed working tension; our margin of only 1.49 would be judged inadequate in professional practice.

- Particle model. The engine is bulky; if it swings, part of the motion is rotational and the simple model no longer describes it exactly.

Conclusion. The model predicts a safe, gentle descent ($a = 1.4\ \text{m s}^{-2}$, $T = 1008\ \text{N}$, landing speed $2.05\ \text{m s}^{-1}$), a counterweight overshoot of about $0.21\ \text{m}$ above its release position, and shows that the same scrap counterweight can haul the engine back up the rough ramp with acceleration about $0.71\ \text{m s}^{-2}$. The critique shows, however, that for real workshop use Mr. Ekane should choose a rope rated well above 1500 N, release the load smoothly, and keep the beam axle well greased.

8 METHODS AND WORKED EXAMPLES

8.1 Skill 1: Setting Up Equations of Motion for Two Connected Particles on a Smooth Surface

Two particles are said to be **connected** when they are joined by a string, rod or tow-bar so that the motion of one constrains the motion of the other. Throughout this chapter we use the standard GCE modelling assumptions: the string is **light** (its mass is negligible) and **inextensible** (its length is constant), and every pulley is **smooth** (no friction in its axle) and light. These assumptions have two crucial consequences:

PROPERTY **Consequences of the modelling assumptions**

For a light inextensible string passing over a smooth pulley:

- the **tension** has the same value T at every point of the string;
- the two particles have accelerations of the same magnitude a (and velocities of the same magnitude), because the length of the string is constant.

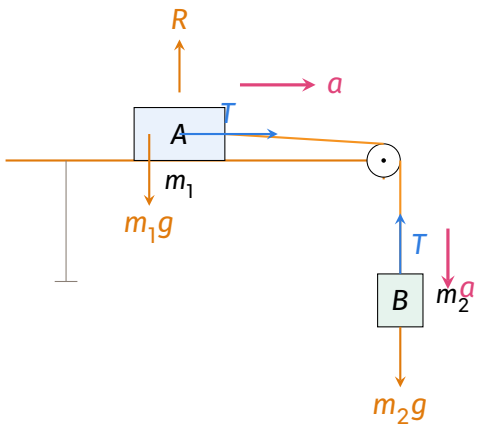
These facts are used in every problem of this chapter; proofs are not examinable, but you must be able to use the results.

METHOD **Two particles connected on a smooth horizontal surface**

To analyse two particles of masses m_1 and m_2 connected by a light inextensible string (possibly passing over a smooth pulley), with no friction:

1. Draw a clear diagram showing each particle separately, with all forces acting on that particle only (weight, normal reaction, tension).
2. Choose a positive direction of motion for each particle, consistent with the string: because the string is inextensible, both particles have the **same magnitude of acceleration** a and the same tension T acts throughout (string light, pulley smooth).
3. Apply Newton's second law $F = ma$ to each particle along its direction of motion.
4. Solve the two simultaneous equations: add them to eliminate T and find a; substitute back to find T.
5. Check: T must be positive and less than the weight of the descending mass; $a < g$ always.

Key reflex: never write the weight of particle 1 in the equation of particle 2. Each particle has its own equation.



Particle A on a smooth table connected over a smooth pulley to the hanging particle B. Each particle carries its own forces; the tension T is the same on both sides.

WORKED EXAMPLE 18 **Car pulling a trailer on a smooth surface**

A particle A of mass 4 kg rests on a smooth horizontal table. It is connected by a light inextensible string, passing over a small smooth pulley fixed at the edge of the table, to a particle B of mass 2 kg hanging freely. The system is released from rest. Taking $g = 9.8\ \text{m s}^{-2}$, find (a) the acceleration of the system, (b) the tension in the string, (c) the speed of B after it has fallen 0.9 m.

SOLUTION

Solution. The table is smooth, so the only horizontal force on A is the tension T. The string is light and inextensible and the pulley smooth, so the tension is the same on both sides and both particles have acceleration of magnitude a.

Equation for A (horizontal motion):

$$T = 4a \quad (1)$$

Equation for B (vertical motion, downwards positive):

$$2g - T = 2a \quad (2)$$

(a) Acceleration. Add (1) and (2) to eliminate T:

$$2g = 6a \quad \Longrightarrow \quad a = \frac{2 \times 9.8}{6} = \frac{19.6}{6} \approx 3.27\ \text{m s}^{-2}.$$

(b) Tension. From (1):

$$T = 4a = 4 \times 3.27 \approx 13.1\ \text{N}.$$

Check: $T = 13.1\ \text{N} < 2g = 19.6\ \text{N}$, as expected since B accelerates downwards.

(c) Speed after falling 0.9 m. The acceleration is constant, so use $v^2 = u^2 + 2as$ with $u = 0$, $s = 0.9$ m:

$$v^2 = 0 + 2 \times 3.27 \times 0.9 = 5.88 \quad \Longrightarrow \quad v \approx 2.42\ \text{m s}^{-1}.$$

WARNING **Frequent mistakes with two connected particles**

- Writing $T = m_2 g$ for the hanging mass. This is only true in equilibrium; when the system accelerates, $T \neq m_2 g$.
- Giving the two particles different accelerations. An inextensible string forces $|a_1| = |a_2|$.
- Mixing the two equations: the mass in $F = ma$ must be the mass of that particle, and the forces must be the forces acting on that particle.

8.2 Skill 2: Motion on a Rough Horizontal Surface

METHOD **Connected particles with friction**

When one particle moves on a **rough** surface:

1. Identify the normal reaction R on the particle on the surface. On a horizontal surface with no vertical component of other forces, $R = mg$.
2. Compute the friction force $F = \mu R$ (motion occurs, so friction is limiting: $F = \mu R$), directed opposite to the motion.
3. Write Newton's second law for each particle, including F in the equation of the particle on the surface.
4. Solve simultaneously for a and T.
5. Check the motion assumption: motion only occurs if the driving force exceeds μR; if a comes out negative, the assumption of motion in that direction was wrong.

Key formula: for mass m_1 on a rough table pulled by hanging mass m_2:

$$a = \frac{(m_2 - \mu m_1)g}{m_1 + m_2}.$$

WORKED EXAMPLE 19 **Rough table problem**

A block P of mass 5 kg lies on a rough horizontal table, the coefficient of friction between the block and the table being 0.2. P is connected by a light inextensible string passing over a smooth pulley at the edge of the table to a block Q of mass 3 kg hanging freely. The system is released from rest. Taking $g = 9.8\ \text{m s}^{-2}$, find (a) the acceleration of the blocks, (b) the tension in the string, (c) the time taken for Q to descend 1.2 m.

SOLUTION

Solution.
Step 1: Reaction and friction on P. Vertically, P is in equilibrium:

$$R = 5g = 5 \times 9.8 = 49\ \text{N}.$$

Since P slides, friction is limiting:

$$F = \mu R = 0.2 \times 49 = 9.8\ \text{N},$$

acting opposite to the motion of P.
Step 2: Equations of motion. Let a be the acceleration and T the tension.
For P (horizontal): $T - F = 5a$, i.e.

$$T - 9.8 = 5a \quad (1)$$

For Q (vertical, downwards positive):

$$3g - T = 3a \quad \Longrightarrow \quad 29.4 - T = 3a \quad (2)$$

Step 3: Solve. Adding (1) and (2):

$$29.4 - 9.8 = 8a \quad \Longrightarrow \quad a = \frac{19.6}{8} = 2.45\ \text{m s}^{-2}.$$

Since $a > 0$, motion does occur as assumed (indeed $3g = 29.4 > F = 9.8$).
From (1): $T = 5a + 9.8 = 12.25 + 9.8 = 22.05$ N.
Step 4: Time to descend 1.2 m. Using $s = ut + \frac{1}{2}at^2$ with $u = 0$:

$$1.2 = \tfrac{1}{2} \times 2.45 \times t^2 \quad \Longrightarrow \quad t^2 = \frac{2.4}{2.45} = 0.9796 \quad \Longrightarrow \quad t \approx 0.99\ \text{s}.$$

WARNING **Friction opposes motion, not the tension**

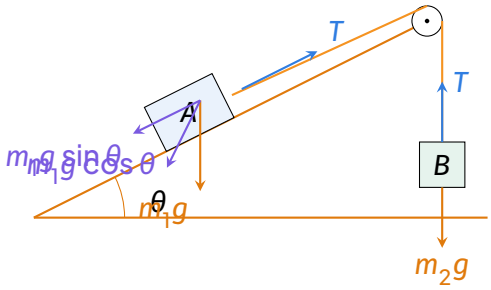
Friction always acts opposite to the direction of sliding (or of impending sliding). Do not automatically place F on the left of your diagram: decide first which way the particle moves, then put F the other way. Also remember $F = \mu R$ is valid only when sliding actually occurs; otherwise $F \le \mu R$.

8.3 Skill 3: Motion on an Inclined Plane

METHOD **Particle on an inclined plane connected to a hanging mass**

1. Resolve the weight of the particle on the plane into components: $mg\sin\theta$ down the plane and $mg\cos\theta$ into the plane.
2. Normal reaction: $R = mg\cos\theta$ (no acceleration perpendicular to the plane).
3. Friction (if rough): $F = \mu R = \mu mg\cos\theta$, opposing the motion (i.e. acting down the plane if the particle moves up, and vice versa).
4. Write $F = ma$ along the plane for the particle on the plane, and vertically for the hanging particle.
5. Solve simultaneously. Decide the direction of motion first by comparing $m_2 g$ with $m_1 g\sin\theta + \mu m_1 g\cos\theta$.

Key reflex: on a smooth plane, $a = \dfrac{(m_2 - m_1\sin\theta)g}{m_1 + m_2}$ when m_2 descends.



Resolving the weight on an inclined plane: component $m_1 g\sin\theta$ down the plane, $m_1 g\cos\theta$ perpendicular into the plane.

WORKED EXAMPLE 20 **Smooth inclined plane — motion does occur**

A particle A of mass 3 kg rests on a smooth plane inclined at 30° to the horizontal. A is connected by a light inextensible string, parallel to the line of greatest slope and passing over a smooth pulley at the top of the plane, to a particle B of mass 5 kg hanging freely. The system is released from rest. Taking $g = 9.8\ \text{m s}^{-2}$, find the acceleration and the tension in the string.

SOLUTION

Solution. Component of A's weight down the plane: $3g\sin 30° = 3 \times 9.8 \times 0.5 = 14.7$ N. Weight of B: $5g = 49$ N. Since $49 > 14.7$ and the plane is smooth, B descends and A moves

up the plane.
Equation for A (up the plane positive):

$$T - 3g\sin 30^\circ = 3a \quad \Longrightarrow \quad T - 14.7 = 3a \quad (1)$$

Equation for B (downwards positive):

$$5g - T = 5a \quad \Longrightarrow \quad 49 - T = 5a \quad (2)$$

Solve. Adding (1) and (2):

$$49 - 14.7 = 8a \quad \Longrightarrow \quad a = \frac{34.3}{8} = 4.29\ \text{m s}^{-2}.$$

From (1): $T = 14.7 + 3 \times 4.29 = 14.7 + 12.86 = 27.56$ N.
Check: $3g\sin 30^\circ = 14.7 < T = 27.56 < 49 = 5g$. ✓

WORKED EXAMPLE 21 **Rough inclined plane — when nothing moves**

A particle A of mass 6 kg rests on a rough plane inclined at 30° to the horizontal, the coefficient of friction being $\frac{1}{4}$. A is connected by a light inextensible string, parallel to the line of greatest slope and passing over a smooth pulley at the top of the plane, to a particle B of mass 4 kg hanging freely. The system is released from rest. Taking $g = 9.8\ \text{m s}^{-2}$, find the acceleration of the system and the tension in the string.

SOLUTION

Solution.
Step 1: Which way does the system move? Component of A's weight down the plane:

$$6g\sin 30^\circ = 6 \times 9.8 \times 0.5 = 29.4\ \text{N}.$$

Weight of B: $4g = 39.2$ N. Since $39.2 > 29.4$ even before friction, we first test the hypothesis that B descends and A moves up the plane; friction on A would then act down the plane.
Step 2: Reaction and friction.

$$R = 6g\cos 30^\circ = 6 \times 9.8 \times \frac{\sqrt{3}}{2} = 29.4\sqrt{3} \approx 50.9\ \text{N},$$

$$F = \mu R = \tfrac{1}{4} \times 50.9 \approx 12.7\ \text{N}.$$

Step 3: Test the hypothesis. For motion with B descending we would need the driving force to exceed the resisting forces:

$$4g \;?\; 6g\sin 30^\circ + F \quad \Longleftrightarrow \quad 39.2 \;?\; 29.4 + 12.7 = 42.1.$$

Since $39.2 < 42.1$, the hanging mass cannot pull A up the plane.

Step 4: Test the opposite direction. Could A slide down, pulling B up? Then friction acts up the plane and we would need

$$6g \sin 30° \;?\; 4g + F \quad \Longleftrightarrow \quad 29.4 \;?\; 39.2 + 12.7 = 51.9,$$

which is also false.
Conclusion: the system remains at rest: $a = 0$. Equilibrium of B gives $T = 4g = 39.2$ N, and equilibrium of A along the plane gives friction $= T - 6g \sin 30° = 39.2 - 29.4 = 9.8$ N, acting down the plane. Since $9.8 < \mu R = 12.7$ N, friction is not limiting, which is consistent with rest.
Lesson: always verify the direction of motion; if the computed acceleration comes out negative, the assumed motion does not occur and friction may not be limiting.

8.4 Skill 4: Atwood's Machine

METHOD **Atwood's machine: two masses over a fixed pulley**

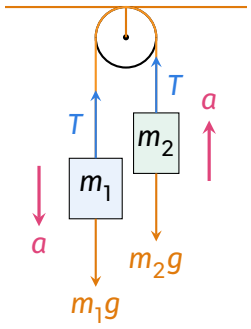
For masses $m_1 > m_2$ hanging on either side of a light inextensible string over a smooth fixed pulley:

1. Take downwards positive for the heavier mass m_1, upwards positive for m_2; both have acceleration of magnitude a.
2. Equations: $m_1 g - T = m_1 a$ and $T - m_2 g = m_2 a$.
3. Add to get the standard results:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}.$$

4. Then use constant-acceleration kinematics ($v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$) for any further question.

Check: $m_2 g < T < m_1 g$; if $m_1 = m_2$ then $a = 0$ and $T = m_1 g$.



Atwood's machine: the heavier mass m_1 accelerates downwards, the lighter mass m_2 upwards, with the same magnitude a; the tension T is the same on both sides.

WORKED EXAMPLE 22 **Classic Atwood's machine**

Particles of masses 5 kg and 3 kg are attached to the ends of a light inextensible string passing over a smooth fixed pulley. The system is released from rest with the string taut and the 5 kg particle 2 m above the ground. Taking $g = 9.8\ \text{m s}^{-2}$, find (a) the acceleration and the tension, (b) the speed of the 5 kg mass when it hits the ground, (c) the time taken.

SOLUTION

Solution.

(a) With $m_1 = 5$, $m_2 = 3$:

$$a = \frac{(5-3)g}{5+3} = \frac{2 \times 9.8}{8} = 2.45\ \text{m s}^{-2},$$

$$T = \frac{2 \times 5 \times 3 \times 9.8}{8} = \frac{294}{8} = 36.75\ \text{N}.$$

Check: $3g = 29.4 < 36.75 < 49 = 5g$. ✓

(b) Using $v^2 = u^2 + 2as$ with $u = 0$, $s = 2$ m:

$$v^2 = 2 \times 2.45 \times 2 = 9.8 \quad \Longrightarrow \quad v = \sqrt{9.8} \approx 3.13\ \text{m s}^{-1}.$$

(c) Using $s = \tfrac{1}{2}at^2$:

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{4}{2.45}} = \sqrt{1.633} \approx 1.28\ \text{s}.$$

DID YOU KNOW? **Why Atwood built this machine**

George Atwood introduced his machine in 1784 to slow down gravity: with nearly equal masses, $a = \dfrac{(m_1 - m_2)g}{m_1 + m_2}$ is a small, easily measured fraction of g, so the laws of accelerated

motion could be checked accurately with the clocks of the time. The same trick — diluting g — is used today in school laboratories with air tracks and ticker-timers.

8.5 Skill 5: Multi-Stage Motion (String Breaks or a Mass is Removed)

METHOD **Multi-stage problems**

When the string breaks, or a mass hits the ground and the string goes slack:

1. Stage 1: analyse the connected motion as usual; find the speed v_1 and positions at the instant the regime changes.
2. Stage 2: after the change, each remaining free particle moves under gravity alone ($a = \pm g$) if the string is slack, or a new connected system forms with new equations. The **initial velocity** of stage 2 is the final velocity of stage 1.
3. Combine the stages to answer questions on total time, greatest height, etc.

Trap: when the string goes slack, the rising mass continues upwards, decelerating at g — it does not stop immediately.

WORKED EXAMPLE 23 **Mass hits the ground; how high does the other rise?**

In an Atwood's machine, masses of 4 kg and 2 kg hang at the same height, 1.5 m above the floor, connected over a smooth fixed pulley. The system is released from rest. The 4 kg mass falls to the floor and does not rebound. Taking $g = 9.8\ \text{m s}^{-2}$, find the greatest height above the floor reached by the 2 kg mass.

SOLUTION

Solution.
Stage 1: connected motion. The acceleration is

$$a = \frac{(4-2)g}{4+2} = \frac{2\times 9.8}{6} = \frac{9.8}{3} \approx 3.267\ \text{m s}^{-2}.$$

When the 4 kg mass has fallen 1.5 m, the common speed is given by $v^2 = 2as$:

$$v_1^2 = 2\times\frac{9.8}{3}\times 1.5 = 9.8 \quad\Longrightarrow\quad v_1 = \sqrt{9.8} \approx 3.13\ \text{m s}^{-1}.$$

At this instant the 2 kg mass is at height $1.5 + 1.5 = 3$ m, moving upwards at $3.13\ \text{m s}^{-1}$.
Stage 2: string slack. Once the 4 kg mass hits the floor, the string goes slack and the 2 kg mass moves freely under gravity, decelerating at g. The extra height gained is given by

$v^2 = u^2 - 2gh$ with final $v = 0$:

$$0 = 9.8 - 2 \times 9.8 \times h \quad \Longrightarrow \quad h = \frac{9.8}{19.6} = 0.5\,\text{m}.$$

Greatest height:

$$H = 3 + 0.5 = 3.5\,\text{m above the floor.}$$

(One must of course assume the pulley is high enough for this to be possible.)

EXERCISE 7 **Two-stage Atwood problem**

Two particles of masses 6 kg and 4 kg are connected by a light inextensible string over a smooth fixed pulley. Initially both are held at rest, 1.8 m above the floor, and then released. The 6 kg particle strikes the floor and does not rebound. Taking $g = 9.8\,\text{m s}^{-2}$, find (a) the speed of the 4 kg particle at the instant the 6 kg particle hits the floor, (b) the greatest height reached by the 4 kg particle, (c) the total time from release until the 4 kg particle reaches its greatest height.

SOLUTION **Exercise 1**

(a) Acceleration: $a = \dfrac{(6-4)g}{6+4} = \dfrac{2 \times 9.8}{10} = 1.96\,\text{m s}^{-2}$. Speed after falling 1.8 m: $v_1^2 = 2 \times 1.96 \times 1.8 = 7.056$, so $v_1 \approx 2.66\,\text{m s}^{-1}$.

(b) Height at that instant: $1.8 + 1.8 = 3.6$ m. Extra rise under gravity: $h = \dfrac{v_1^2}{2g} = \dfrac{7.056}{19.6} = 0.36$ m. Greatest height: $H = 3.6 + 0.36 = 3.96$ m.

(c) Stage 1 time: $t_1 = \dfrac{v_1}{a} = \dfrac{2.66}{1.96} \approx 1.36$ s. Stage 2 time: $t_2 = \dfrac{v_1}{g} = \dfrac{2.66}{9.8} \approx 0.27$ s. Total: $t \approx 1.63$ s.

8.6 Skill 6: Pulley Systems with More Than Two Masses

METHOD **Three or more connected masses**

1. Draw each mass separately and mark the tension in each separate string: different strings generally have different tensions $T_1, T_2, \ldots$
2. Use inextensibility to relate the accelerations. In the common GCE case (masses in series on a table, or one mass on a table with two hanging masses on separate strings), all accelerations have the same magnitude a.

3. Write $F = ma$ for every mass — one equation per mass.
4. Solve the system: adding all equations eliminates all internal tensions at once and gives a directly; then back-substitute for each tension.

Key reflex: internal tensions cancel when you add all the equations — the "driving force" is the sum of external forces along the direction of motion, and the "total mass" is the sum of all masses.

WORKED EXAMPLE 24 **Two hanging masses, one on a smooth table**

A particle A of mass 3 kg lies on a smooth horizontal table. A light inextensible string connects A over a smooth pulley at one edge of the table to a hanging mass B of 2 kg; a second light inextensible string connects A over a smooth pulley at the opposite edge to a hanging mass C of 5 kg. The system is released from rest. Taking $g = 9.8\ \text{m s}^{-2}$, find the acceleration of the system and the tension in each string.

SOLUTION

Solution. Since $5 > 2$, mass C descends, B rises, and A moves towards C's edge. Let T_1 be the tension in the string joining A to B, and T_2 the tension in the string joining A to C. All accelerations have magnitude a (strings inextensible).
Equation for C (downwards positive):

$$5g - T_2 = 5a \quad (1)$$

Equation for A (towards C positive): the table is smooth, so

$$T_2 - T_1 = 3a \quad (2)$$

Equation for B (upwards positive):

$$T_1 - 2g = 2a \quad (3)$$

Solve. Adding (1), (2) and (3) eliminates both tensions:

$$5g - 2g = (5 + 3 + 2)a \implies 3g = 10a \implies a = \frac{3 \times 9.8}{10} = 2.94\ \text{m s}^{-2}.$$

From (3): $T_1 = 2g + 2a = 19.6 + 5.88 = 25.48$ N.
From (1): $T_2 = 5g - 5a = 49 - 14.7 = 34.3$ N.
Check with (2): $T_2 - T_1 = 34.3 - 25.48 = 8.82 = 3 \times 2.94 = 3a$. ✓
Note $T_1 \neq T_2$: the two strings are different, so their tensions differ.

WARNING **One string, one tension — two strings, two tensions**

The rule "same tension throughout" applies to one light string over smooth pulleys. As soon as the system contains two distinct strings, you must introduce two distinct unknowns T_1 and T_2. Writing a single T everywhere in a two-string system is one of the most common errors at the GCE.

Variable acceleration in connected systems.

In some problems the driving force varies with position or time (for example an elastic rope, or a force $F(t)$ applied to one mass). The method is unchanged in spirit: write Newton's second law for each particle, eliminate the tension, and obtain a as a function of x or t. Then, instead of the constant-acceleration formulae, use

$$a = \frac{dv}{dt} \qquad \text{or} \qquad a = v\,\frac{dv}{dx},$$

and integrate. For instance, if after eliminating T you obtain $a = 4 - 0.5\,x$ (in SI units) for a system starting from rest at $x = 0$, then

$$v\,\frac{dv}{dx} = 4 - 0.5x \quad \Longrightarrow \quad \tfrac{1}{2}v^2 = 4x - 0.25x^2,$$

which gives the speed at any position. The constant-acceleration formulae $v = u + at$, $s = ut + \tfrac{1}{2}at^2$ must **not** be used when a is not constant.

8.7 Skill 7: Energy Methods for Connected Particles

METHOD **Work–energy approach**

For connected systems, energy methods often give speeds much faster than Newton's second law:

1. Identify the displacement s of each mass (equal magnitudes for a single inextensible string).
2. Write the energy balance:

   $$\text{loss of GPE} = \text{gain of KE} + \text{work done against friction}.$$

   The tension does no net work on the whole system (it is internal), so it disappears — this is the great advantage of the method.
3. KE of the system: $E_k = \tfrac{1}{2}(m_1 + m_2)v^2$ since both masses have the same speed v.
4. Friction term: $\mu R\,s = \mu mg\cos\theta \cdot s$ for a mass sliding s along a rough plane.
5. Solve for v. If the acceleration is needed afterwards, use $v^2 = 2as$ (constant a) or differentiate the energy equation.

When to use it: whenever the question asks for a speed after a given distance, and especially for real-life problems (lifts, vehicles, loads on ramps).

WORKED EXAMPLE 25 **Energy on a rough slope — a loading ramp**

At a warehouse in Douala, a crate of mass 50 kg is pulled up a rough ramp inclined at 20° to the horizontal (coefficient of friction 0.3) by a light inextensible rope parallel to the ramp, passing over a smooth pulley at the top, with a counterweight of mass 40 kg hanging vertically. The system is released from rest. Taking $g = 9.8\ \text{m s}^{-2}$, use energy considerations to find the speed of the crate after it has moved 2 m up the ramp, and verify the result by computing the acceleration from Newton's second law.

SOLUTION

Solution.
Step 1: Energy balance over $s = 2$ m.
The counterweight descends 2 m: loss of GPE $= 40 \times 9.8 \times 2 = 784$ J.
The crate rises vertically by $2\sin 20°$: gain of GPE

$$= 50 \times 9.8 \times 2\sin 20° = 980 \times 0.342 = 335.2\ \text{J}.$$

Work done against friction on the crate: $R = 50g\cos 20° = 490 \times 0.9397 = 460.4$ N, so

$$W_F = \mu R s = 0.3 \times 460.4 \times 2 = 276.3\ \text{J}.$$

Gain of kinetic energy of the whole system (both masses have speed v):

$$E_k = \tfrac{1}{2}(50 + 40)v^2 = 45v^2.$$

Step 2: Apply the energy equation.

$$784 = 335.2 + 276.3 + 45v^2$$

$$45v^2 = 172.5 \quad\Longrightarrow\quad v^2 = 3.834 \quad\Longrightarrow\quad v \approx 1.96\ \text{m s}^{-1}.$$

Step 3: Verification by Newton's second law.
For the counterweight: $40g - T = 40a$, i.e. $392 - T = 40a$.
For the crate along the ramp: $T - 50g\sin 20° - \mu(50g\cos 20°) = 50a$, i.e.

$$T - 167.6 - 138.1 = 50a.$$

Adding: $392 - 305.7 = 90a$, so $a = \dfrac{86.3}{90} = 0.959\ \text{m s}^{-2}$.
Then $v^2 = 2as = 2 \times 0.959 \times 2 = 3.836$, giving $v \approx 1.96\ \text{m s}^{-1}$ — the same result. ✓
Remark: the energy method never required the tension; it is shorter and less error-prone when only a speed is required.

EXERCISE 8 **Energy check on Atwood's machine**

In an Atwood's machine with masses 5 kg and 3 kg released from rest, use the conservation of mechanical energy to find the common speed after the 5 kg mass has descended 2 m. Take $g = 9.8\,\text{m s}^{-2}$. Compare with the answer obtained from $a = \dfrac{(m_1 - m_2)g}{m_1 + m_2}$ and $v^2 = 2as$.

SOLUTION **Exercise 2**

Loss of GPE of the 5 kg mass: $5 \times 9.8 \times 2 = 98$ J. Gain of GPE of the 3 kg mass: $3 \times 9.8 \times 2 = 58.8$ J. Net loss $= 98 - 58.8 = 39.2$ J, converted into kinetic energy of both masses:

$$\tfrac{1}{2}(5+3)v^2 = 39.2 \implies 4v^2 = 39.2 \implies v^2 = 9.8 \implies v \approx 3.13\,\text{m s}^{-1}.$$

By dynamics: $a = \dfrac{2 \times 9.8}{8} = 2.45\,\text{m s}^{-2}$, $v^2 = 2 \times 2.45 \times 2 = 9.8$. Same result. ✓

KEY POINTS **Chapter essentials**

- Light inextensible string over a smooth pulley: same tension throughout, same magnitude of acceleration for both particles.
- One equation $F = ma$ per particle; add the equations to eliminate the tension.
- Rough surface: $R = mg$ (horizontal) or $R = mg\cos\theta$ (inclined plane); sliding friction $F = \mu R$ opposes the motion.
- Atwood's machine: $a = \dfrac{(m_1 - m_2)g}{m_1 + m_2}$, $T = \dfrac{2m_1 m_2 g}{m_1 + m_2}$.
- Always check the direction of motion: a negative acceleration means the assumed motion does not occur and friction may not be limiting.
- Multi-stage motion: the final velocity of one stage is the initial velocity of the next; after the string goes slack, particles move under gravity alone.
- Distinct strings have distinct tensions; adding all equations of motion eliminates every internal tension at once.
- Energy method: loss of GPE = gain of KE + work against friction; the tension disappears, giving speeds directly.

9 EXERCISES

9.1 I check my knowledge

EXERCISE 9 **Multiple choice questions**

For each question, choose the single correct answer. Take $g = 9.8\ \text{m s}^{-2}$.

1. Two particles of masses 2 kg and 3 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood's machine). The acceleration of the system is:

   (a) $1.96\ \text{m s}^{-2}$

   (b) $4.9\ \text{m s}^{-2}$

   (c) $9.8\ \text{m s}^{-2}$

   (d) $0.98\ \text{m s}^{-2}$

2. A light inextensible string connects two particles over a smooth pulley. Which statement is correct?

   (a) The tensions on the two sides of the pulley are different.

   (b) The tension is the same throughout the string.

   (c) The string can push as well as pull.

   (d) The acceleration of the two particles may differ in magnitude.

3. A particle rests on a rough horizontal plane and is pulled by a horizontal string. Motion is about to begin. The friction force equals:

   (a) μR

   (b) μmg always

   (c) zero

   (d) mg

4. A car tows a trailer along a horizontal road at constant speed. The tension in the towbar is:

   (a) zero

   (b) equal to the resistance on the trailer

   (c) equal to the driving force

   (d) equal to the total weight

EXERCISE 10 **True or false?**

State whether each statement is true or false, justifying your answer in one or two sentences.

1. In a system of two particles joined by a light inextensible string over a smooth pulley, the magnitudes of the accelerations of the two particles are equal.
2. The tension in a string can be negative.
3. On a rough plane, the friction force is always equal to μR.
4. When a hanging particle in a connected system strikes the ground, the other particle stops immediately.

EXERCISE 11 **Key definitions**

1. Explain what is meant by a light inextensible string and state two consequences of this model for the tension and the accelerations of the connected particles.
2. Explain what is meant by a smooth pulley and state the consequence for the tensions on either side of it.
3. Define the coefficient of friction μ between two surfaces and state the inequality satisfied by the friction force F when the system is in equilibrium.

EXERCISE 12 **Direct calculations**

Take $g = 9.8\ \text{m s}^{-2}$.

1. A particle of mass $5\ \text{kg}$ hangs freely from a light inextensible string and moves upwards with acceleration $1.2\ \text{m s}^{-2}$. Find the tension in the string.
2. A block of mass $8\ \text{kg}$ is pulled along a rough horizontal table ($\mu = 0.3$) by a horizontal force of $40\ \text{N}$. Find the friction force and the acceleration of the block.
3. A particle of mass $4\ \text{kg}$ rests on a smooth plane inclined at 30° to the horizontal, held by a string parallel to the plane. Find the tension in the string and the normal reaction.

9.2 I practise

EXERCISE 13 **Particles on a smooth table**

A particle of mass 4 kg rests on a smooth horizontal table, connected by a light inextensible string passing over a smooth pulley at the edge of the table to a particle of mass 6 kg hanging freely. The system is released from rest. Take $g = 9.8\ \text{m s}^{-2}$.

1. Draw a clear force diagram for each particle.
2. Find the acceleration of the system.
3. Find the tension in the string.
4. Find the magnitude of the resultant force exerted by the string on the pulley.

EXERCISE 14 **Particles on a rough table**

Repeat the previous exercise, but now the table is rough with coefficient of friction $\mu = 0.25$. Take $g = 9.8\ \text{m s}^{-2}$.

1. Show that motion does occur when the system is released.
2. Find the new acceleration and the new tension in the string.
3. Find the speed of the 6 kg mass after it has fallen 0.8 m from rest.

EXERCISE 15 **Masses on an inclined plane**

A mass of 5 kg lies on a plane inclined at 30° to the horizontal, connected by a light inextensible string parallel to the plane, passing over a smooth pulley at the top of the plane, to a hanging mass of 7 kg. Take $g = 9.8\ \text{m s}^{-2}$.

1. Given that the plane is smooth, find the acceleration of the system and the tension in the string.
2. Now suppose the plane is rough with $\mu = 0.2$. Verify the assumed direction of motion, then find the new acceleration and tension.

EXERCISE 16 **Two masses on a table and one hanging**

Two masses of 2 kg and 3 kg rest on a smooth horizontal table, joined by a light inextensible string. The 3 kg mass is connected by a second light inextensible string passing over a smooth pulley at the edge of the table to a hanging mass of 5 kg. Take $g = 9.8\ \text{m s}^{-2}$.

1. Draw the force diagram for each of the three masses.

2. Find the acceleration of the system.
3. Find the tensions T_1 (string between the 2 kg and 3 kg masses) and T_2 (string over the pulley).

9.3 I prepare for the GCE

EXERCISE 17 **Atwood's machine and multi-stage motion** **(14 marks)**

An Atwood's machine consists of particles of masses 3 kg and 5 kg connected by a light inextensible string passing over a smooth fixed pulley. The system is released from rest with both particles at the same height. Take $g = 9.8\,\text{m s}^{-2}$.

(a) Find the acceleration of the system and the tension in the string. (4 marks)

(b) Find the reaction force exerted on the support of the pulley. (2 marks)

(c) After the 5 kg mass has fallen 2 m, it strikes the ground and does not rebound. Find the speed of the 3 kg mass at this instant. (3 marks)

(d) Find the greatest height above its initial position reached by the 3 kg mass, assuming it does not reach the pulley. (5 marks)

EXERCISE 18 **Car and trailer on a hill** **(15 marks)**

A car of mass 1200 kg tows a trailer of mass 400 kg up a hill inclined at an angle $\sin^{-1}\left(\frac{1}{14}\right)$ to the horizontal. Resistance forces of 300 N on the car and 100 N on the trailer act against the motion. The driving force of the engine is 2600 N. Take $g = 9.8\,\text{m s}^{-2}$.

(a) Draw a force diagram for the car and for the trailer. (3 marks)

(b) By considering the car and the trailer separately (or the whole system), find the acceleration. (4 marks)

(c) Find the tension in the tow-bar. (3 marks)

(d) The car now descends the same hill and brakes with a braking force of 1800 N, the resistances being unchanged. Find the thrust (compression) in the tow-bar. (5 marks)

EXERCISE 19 **Rough inclined plane, equilibrium range and energy** **(18 marks)**

A block A of mass 6 kg rests on a rough plane inclined at $\alpha = 25^\circ$ to the horizontal, with coefficient of friction $\mu = 0.4$. The block is connected by a light inextensible string, parallel to the plane and passing over a smooth pulley at the top, to a particle B of mass m hanging freely. Take $g = 9.8\ \text{m s}^{-2}$.

(a) Find the range of values of m for which the system remains at rest. (6 marks)

(b) For $m = 10$ kg, find the acceleration of the system and the tension in the string. (4 marks)

(c) Using the work–energy principle, find the speed of the particles after B has descended 1 m from rest. (5 marks)

(d) Discuss briefly how friction in the pulley bearing would affect your answers to parts (b) and (c). (3 marks)

EXERCISE 20 **A uniform chain sliding off a table** **(14 marks)**

A uniform chain of length 2 m and mass 4 kg lies in a straight line on a smooth horizontal table, perpendicular to the edge, with 0.5 m hanging over the edge. The chain is released from rest. Take $g = 9.8\ \text{m s}^{-2}$.

(a) Show that, when a length x of the chain hangs over the edge, the acceleration of the chain is $a = \dfrac{gx}{2}$. (4 marks)

(b) By writing $a = v\dfrac{dv}{dx}$ and integrating, find the speed of the chain at the instant it just leaves the table. (6 marks)

(c) Verify your answer to part (b) using the conservation of mechanical energy, taking the table top as the reference level for potential energy. (4 marks)

10 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice questions**

1. For an Atwood's machine, applying Newton's second law to each mass and adding gives $a = \dfrac{m_2 - m_1}{m_1 + m_2} g = \dfrac{3-2}{2+3} \times 9.8 = \dfrac{1}{5} \times 9.8 = 1.96\ \text{m s}^{-2}$. Answer: (a).

2. For a **light inextensible string** passing over a **smooth pulley**, the tension is the same throughout the string. A string can only pull (never push), and because the string is inextensible the magnitudes of the accelerations of the two particles are equal. Answer: (b).

3. When motion is about to begin (**limiting equilibrium**), the friction force attains its maximum value $F_{\max} = \mu R$. On a horizontal plane with no vertical acceleration, $R = mg$, so $F_{\max} = \mu mg$. Answer: (a).

4. At constant speed the acceleration is zero, so the resultant force on the trailer is zero. Horizontally the only forces on the trailer are the tension T in the tow-bar and the resistance; hence T equals the resistance force on the trailer. Answer: (b).

SOLUTION **Exercise 2 — True or false?**

1. True. The string is inextensible, so the length of string between the particles is fixed. Differentiating twice, the magnitudes of their velocities and accelerations along the string are equal (their directions in space may differ, e.g. one horizontal, one vertical).

2. False. Tension is a pulling force; a string cannot push. In the model of a light inextensible string the tension always satisfies $T \geq 0$. A negative value would correspond to compression, which a string cannot sustain — it would simply go slack.

3. False. The friction force satisfies $F \leq \mu R$. Equality $F = \mu R$ holds only when sliding occurs (kinetic friction) or at limiting equilibrium (maximum static friction). In static equilibrium below the limit, F takes exactly the value needed to prevent motion.

4. False. When the hanging particle strikes the ground, the string goes slack and the tension vanishes. The other particle continues with the velocity it had at that instant, decelerating only under the remaining forces (friction, or a component of gravity on a slope). It stops immediately only if an impulsive force acts on it.

SOLUTION **Exercise 3 — Key definitions**

1. A light inextensible string is a string whose mass is negligible compared with the masses it connects, and whose length does not change under tension. Consequences:
   - Because it is light, the tension is the same at every point of the string (provided any pulley it passes over is smooth).
   - Because it is inextensible, the magnitudes of the velocities and of the accelerations of the connected particles, measured along the string, are equal.
2. A smooth pulley is a pulley with a frictionless axle and frictionless contact with the string. The consequence is that the tension in the string has the same magnitude on both sides of the pulley.
3. The coefficient of friction μ between two surfaces is the ratio of the limiting friction force to the normal reaction: $\mu = \dfrac{F_{\max}}{R}$. While the surfaces are at rest relative to each other, the friction force satisfies $F \le \mu R$; during sliding, $F = \mu R$ and opposes the relative motion.

SOLUTION **Exercise 4 — Direct calculations**

Take $g = 9.8\ \text{m s}^{-2}$.

1. Forces on the 5 kg particle: tension T upward, weight $5g$ downward. Acceleration $a = 1.2\ \text{m s}^{-2}$ upward. Newton's second law (upward positive):

$$T - 5g = 5a \implies T = 5(g + a) = 5(9.8 + 1.2) = 5 \times 11 = 55\ \text{N}.$$

2. Block of mass 8 kg, $\mu = 0.3$, horizontal pull $P = 40$ N. Normal reaction: $R = 8g = 78.4$ N. Maximum (limiting) friction: $F_{\max} = \mu R = 0.3 \times 78.4 = 23.52$ N. Since $P = 40\ \text{N} > F_{\max}$, the block slides. Kinetic friction: $F = 23.52$ N. Resultant force: $40 - 23.52 = 16.48$ N. Acceleration: $a = \dfrac{16.48}{8} = 2.06\ \text{m s}^{-2}$.
3. Mass 4 kg on a smooth plane inclined at 30°, held at rest by a string parallel to the plane. Resolve perpendicular to the plane: $R = 4g\cos 30^\circ = 4 \times 9.8 \times \dfrac{\sqrt{3}}{2} = 19.6\sqrt{3} \approx 33.9$ N. Resolve parallel to the plane (equilibrium): $T - 4g\sin 30^\circ = 0 \implies T = 4g\sin 30^\circ = 4 \times 9.8 \times 0.5 = 19.6$ N.

SOLUTION **Exercise 5 — Particles on a smooth table**

Masses: $m_1 = 4\,\text{kg}$ on the table, $m_2 = 6\,\text{kg}$ hanging. Smooth table, smooth pulley, light inextensible string. $g = 9.8\,\text{m s}^{-2}$.

1. Force diagrams:
   - 4 kg particle: weight $4g$ vertically downward, normal reaction R vertically upward, tension T horizontal towards the pulley.
   - 6 kg particle: weight $6g$ downward, tension T upward.
2. Equations of motion (both particles have acceleration of magnitude a): For the 4 kg particle (horizontal): $T = 4a$. For the 6 kg particle (downward positive): $6g - T = 6a$. Adding eliminates T: $6g = 10a \implies a = \dfrac{6 \times 9.8}{10} = 5.88\,\text{m s}^{-2}$.
3. Tension: $T = 4a = 4 \times 5.88 = 23.52\,\text{N}$.
4. The string exerts two forces of magnitude T on the pulley: one horizontal (towards the table) and one vertically downward. These are perpendicular, so the resultant has magnitude
$$\sqrt{T^2 + T^2} = T\sqrt{2} = 23.52\sqrt{2} \approx 33.3\,\text{N},$$
acting at 45° below the horizontal, directed towards the edge of the table.

SOLUTION **Exercise 6 — Particles on a rough table**

Same arrangement as Exercise 5, but the table is rough with $\mu = 0.25$. $g = 9.8\,\text{m s}^{-2}$.

1. Does motion occur? Normal reaction on the 4 kg block: $R = 4g = 39.2\,\text{N}$. Maximum available friction: $F_{\max} = \mu R = 0.25 \times 39.2 = 9.8\,\text{N}$. If the system were held in equilibrium, the string would have to transmit the full weight of the hanging mass, $6g = 58.8\,\text{N}$, to the block. Since $58.8 > 9.8$, friction cannot prevent motion: the system moves.
2. With sliding, friction takes its limiting value $F = 9.8\,\text{N}$, opposing the motion of the block. 4 kg block: $T - 9.8 = 4a$. 6 kg mass: $58.8 - T = 6a$. Adding: $58.8 - 9.8 = 10a \implies 49 = 10a \implies a = 4.9\,\text{m s}^{-2}$. Tension: $T = 4a + 9.8 = 19.6 + 9.8 = 29.4\,\text{N}$.
3. Speed after the hanging mass has fallen 0.8 m from rest:
$$v^2 = u^2 + 2as = 0 + 2 \times 4.9 \times 0.8 = 7.84 \implies v = \sqrt{7.84} = 2.8\,\text{m s}^{-1}.$$

SOLUTION **Exercise 7 — Masses on an inclined plane**

$m_A = 5\,\text{kg}$ on a plane inclined at $\alpha = 30^\circ$, connected by a string parallel to the plane, over a smooth pulley at the top, to $m_B = 7\,\text{kg}$ hanging freely. $g = 9.8\,\text{m s}^{-2}$.

1. Smooth plane. Component of A's weight down the plane: $5g \sin 30^\circ = 24.5\,\text{N}$. Since

$7g = 68.6 > 24.5$, assume B descends and A moves up the plane with acceleration of magnitude a. A (up the plane): $T - 24.5 = 5a$. B (downward): $68.6 - T = 7a$. Adding: $68.6 - 24.5 = 12a \implies 44.1 = 12a \implies a = 3.675\,\text{m s}^{-2}$. Tension: $T = 24.5 + 5 \times 3.675 = 24.5 + 18.375 = 42.875\,\text{N}$.

2. Rough plane, $\mu = 0.2$. Normal reaction on A: $R = 5g\cos 30^\circ = 5 \times 9.8 \times \dfrac{\sqrt{3}}{2} \approx 42.44\,\text{N}$. Maximum friction: $F_{\max} = \mu R = 0.2 \times 42.44 \approx 8.49\,\text{N}$. Direction of motion: if A moves up the plane, the forces resisting this are $24.5 + 8.49 = 32.99\,\text{N}$. Since $7g = 68.6 > 32.99$, B does descend and A moves up the plane; friction on A acts down the plane. Equations: A: $T - 5g\sin 30^\circ - \mu(5g\cos 30^\circ) = 5a$, i.e. $T - 24.5 - 8.49 = 5a$. B: $68.6 - T = 7a$. Adding: $68.6 - 24.5 - 8.49 = 12a \implies 35.61 = 12a \implies a \approx 2.97\,\text{m s}^{-2}$. Tension: $T = 68.6 - 7 \times 2.97 \approx 68.6 - 20.77 \approx 47.8\,\text{N}$.

SOLUTION **Exercise 8 — Two masses on a table and one hanging**

$m_1 = 2\,\text{kg}$ and $m_2 = 3\,\text{kg}$ on a smooth table, joined by a light inextensible string; m_2 is joined by a second string passing over a smooth pulley at the edge to $m_3 = 5\,\text{kg}$ hanging freely. $g = 9.8\,\text{m s}^{-2}$.

1. Force diagrams (horizontal forces only for the table masses):

   - $2\,\text{kg}$: tension T_1 pulling towards m_2 (the only horizontal force).
   - $3\,\text{kg}$: tension T_1 pulling back towards m_1, tension T_2 pulling towards the pulley.
   - $5\,\text{kg}$: weight $5g$ downward, tension T_2 upward.

   Note that the two strings are different, so $T_1 \neq T_2$ in general.

2. All three masses move with acceleration of magnitude a (both strings inextensible): $2\,\text{kg}$: $T_1 = 2a$. $3\,\text{kg}$: $T_2 - T_1 = 3a$. $5\,\text{kg}$: $5g - T_2 = 5a$. Adding all three equations eliminates both tensions:

$$5g = (2 + 3 + 5)a = 10a \implies a = \frac{5 \times 9.8}{10} = 4.9\,\text{m s}^{-2}.$$

3. $T_1 = 2a = 9.8\,\text{N}$; $T_2 = 5g - 5a = 49 - 24.5 = 24.5\,\text{N}$. Check with the middle equation: $T_2 - T_1 = 24.5 - 9.8 = 14.7 = 3 \times 4.9$ ✓.

SOLUTION **Exercise 9 — Atwood's machine and multi-stage motion** **(14 marks)**

Masses $3\,\text{kg}$ and $5\,\text{kg}$ connected by a light inextensible string over a smooth pulley; released from rest at the same height. $g = 9.8\,\text{m s}^{-2}$.

(a) Acceleration and tension. (4 marks) Let a be the acceleration of the $5\,\text{kg}$ mass down-

ward (and of the 3 kg mass upward). 5 kg: $5g - T = 5a$. 3 kg: $T - 3g = 3a$. Adding: $2g = 8a \implies a = \frac{g}{4} = \frac{9.8}{4} = 2.45\ \text{m s}^{-2}$. $T = 3g + 3a = 3(g + a) = 3(9.8 + 2.45) = 3 \times 12.25 = 36.75$ N. Mark scheme: M1 for both equations of motion, A1 for $a = 2.45$, M1 for substituting back, A1 for $T = 36.75$ N.

(b) Reaction on the pulley support. (2 marks) The string exerts two downward forces, each of magnitude T, on the pulley. Since the pulley is light and in equilibrium, the support reaction is

$$R = 2T = 2 \times 36.75 = 73.5\ \text{N}.$$

Mark scheme: M1 for $R = 2T$, A1 for 73.5 N.

(c) Speed when the 5 kg mass hits the ground after falling 2 m. (3 marks)

$$v^2 = u^2 + 2as = 0 + 2 \times 2.45 \times 2 = 9.8 \implies v = \sqrt{9.8} \approx 3.13\ \text{m s}^{-1}.$$

Both particles have this speed at that instant (inextensible string). Mark scheme: M1 for $v^2 = u^2 + 2as$, A1 for $v^2 = 9.8$, A1 for $v \approx 3.13\ \text{m s}^{-1}$.

(d) Greatest height reached by the 3 kg mass above its initial position. (5 marks) While the string is taut, the 3 kg mass rises 2 m (the same distance the 5 kg mass falls). When the 5 kg mass hits the ground, the string goes slack ($T = 0$). The 3 kg mass continues upward as a free projectile with initial speed $u = \sqrt{9.8}\ \text{m s}^{-1}$ under gravity alone. Extra height gained: $0 = u^2 - 2gh \implies h = \frac{u^2}{2g} = \frac{9.8}{2 \times 9.8} = 0.5$ m. Greatest height above the initial position: $2 + 0.5 = 2.5$ m. Mark scheme: M1 for recognising the string goes slack, M1 for $v^2 = u^2 - 2gh$, A1 for $h = 0.5$ m, M1 for adding the 2 m already gained, A1 for 2.5 m.

SOLUTION **Exercise 10 — Car and trailer on a hill** **(15 marks)**

Car $m_C = 1200$ kg, trailer $m_T = 400$ kg, hill inclined at θ with $\sin\theta = \frac{1}{14}$. Resistances: 300 N on the car, 100 N on the trailer. Driving force $F = 2600$ N acting up the hill. $g = 9.8\ \text{m s}^{-2}$. Useful values: $1200g\sin\theta = 1200 \times 9.8 \times \frac{1}{14} = 840$ N; $400g\sin\theta = 400 \times 9.8 \times \frac{1}{14} = 280$ N.

(a) Force diagrams. (3 marks)

- Car: weight $1200g$ vertically down, normal reaction perpendicular to the plane, driving force 2600 N up the plane, resistance 300 N down the plane, tension T in the tow-bar down the plane (the trailer pulls back on the car).
- Trailer: weight $400g$ vertically down, normal reaction perpendicular to the plane, resistance 100 N down the plane, tension T up the plane (the car pulls the trailer forward).

(b) Acceleration. (4 marks) Treat the car and trailer as a single system of mass 1600 kg (the

tension is internal and cancels):

$$F - 300 - 100 - (1200 + 400)g\sin\theta = 1600a$$

$$2600 - 400 - 1120 = 1600a \implies 1080 = 1600a \implies a = 0.675\ \text{m s}^{-2}.$$

Mark scheme: M1 for resolving the weights along the plane, M1 for the resultant-force equation, A1 for 1080 N, A1 for $a = 0.675\ \text{m s}^{-2}$.

(c) Tension in the tow-bar. (3 marks) Apply Newton's second law to the trailer alone (up the plane positive):

$$T - 100 - 400g\sin\theta = 400a \implies T - 100 - 280 = 400 \times 0.675$$

$$T = 380 + 270 = 650\ \text{N}.$$

Mark scheme: M1 for the trailer equation, A1 for $400g\sin\theta = 280$ N, A1 for $T = 650$ N.

(d) Thrust in the tow-bar when descending with braking force 1800 N. (5 marks) The vehicle now moves down the plane, so the braking force (1800 N on the car) and both resistances act up the plane, while the weight components act down the plane. Let P be the thrust (compression) in the tow-bar: the car pushes the trailer down the plane and the trailer pushes the car up the plane. Take down the plane as positive; let the acceleration be a. Car: $840 - 300 - 1800 + P = 1200a \implies P - 1260 = 1200a$. Trailer: $280 - 100 - P = 400a \implies 180 - P = 400a$. Adding: $-1080 = 1600a \implies a = -0.675\ \text{m s}^{-2}$. The negative sign means the acceleration is directed up the plane: the vehicle is decelerating as it descends, which is consistent with braking. Then $P = 180 - 400a = 180 - 400(-0.675) = 180 + 270 = 450$ N. The thrust in the tow-bar is 450 N. Mark scheme: M1 for correct force directions, M1 for both equations, A1 for $a = -0.675$, M1 for solving for P, A1 for $P = 450$ N.

SOLUTION **Exercise 11 — Rough inclined plane, equilibrium range and energy (18 marks)**

Block A of mass 6 kg on a rough plane inclined at $\alpha = 25^\circ$, $\mu = 0.4$; string parallel to the plane over a smooth pulley to a hanging mass B of mass m kg. $g = 9.8\ \text{m s}^{-2}$. Preliminary values: $\sin 25^\circ \approx 0.4226$, $\cos 25^\circ \approx 0.9063$. $6g\sin 25^\circ \approx 6 \times 9.8 \times 0.4226 \approx 24.85$ N; $R = 6g\cos 25^\circ \approx 53.29$ N; $F_{\max} = \mu R = 0.4 \times 53.29 \approx 21.32$ N.

(a) Range of m for equilibrium. (6 marks) In equilibrium the tension equals the weight of B: $T = mg$. Friction on A can act either way along the plane, up to $F_{\max}$. The two limiting cases are:

- A on the point of sliding down the plane (m smallest): friction acts up the plane at its maximum. $T + F_{\max} = 6g\sin 25^\circ \implies m_{\min}g = 6g\sin 25^\circ - \mu(6g\cos 25^\circ)$. $m_{\min} = 6(\sin 25^\circ - 0.4\cos 25^\circ) = 6(0.4226 - 0.3625) = 6 \times 0.0601 \approx 0.361$ kg.
- A on the point of sliding up the plane (m largest): friction acts down the plane at

its maximum. $T = 6g\sin 25^\circ + F_{\max} \Longrightarrow m_{\max} g = 6g\sin 25^\circ + \mu(6g\cos 25^\circ)$. $m_{\max} = 6(\sin 25^\circ + 0.4\cos 25^\circ) = 6(0.4226 + 0.3625) = 6 \times 0.7851 \approx 4.71\,\text{kg}$.

Equilibrium is possible for $0.361\,\text{kg} \le m \le 4.71\,\text{kg}$. Mark scheme: B1 for $R = 6g\cos 25^\circ$, B1 for $F_{\max} = \mu R$, M1 for the two limiting equations, A1 for $m_{\min}$, A1 for $m_{\max}$, A1 for the range.

(b) For $m = 10\,\text{kg}$, acceleration and tension. (4 marks) Since $10 > m_{\max}$, B descends and A slides up the plane; kinetic friction $F = 21.32\,\text{N}$ acts down the plane on A. A (up the plane): $T - 24.85 - 21.32 = 6a$. B (downward): $10g - T = 10a$, i.e. $98 - T = 10a$. Adding: $98 - 24.85 - 21.32 = 16a \Longrightarrow 51.83 = 16a \Longrightarrow a \approx 3.24\,\text{m s}^{-2}$. $T = 98 - 10a = 98 - 32.40 \approx 65.6\,\text{N}$. Mark scheme: M1 for both equations, A1 for $a \approx 3.24\,\text{m s}^{-2}$, M1 for substituting back, A1 for $T \approx 65.6\,\text{N}$.

(c) Work–energy principle: speed after B has descended 1 m. (5 marks) While B descends 1 m, A moves 1 m up the plane (inextensible string). Both masses have the same speed v. Work done by gravity on B: $+10g \times 1 = +98$ J. Work done by gravity on A: $-6g\sin 25^\circ \times 1 = -24.85$ J. Work done by friction on A: $-21.32 \times 1 = -21.32$ J. (The tension does no net work on the whole system: it does $+T$ on A and $-T$ on B.) Total work = gain in kinetic energy of both masses:

$$98 - 24.85 - 21.32 = \tfrac{1}{2}(6 + 10)v^2 \Longrightarrow 51.83 = 8v^2$$

$$v^2 = 6.479 \Longrightarrow v \approx 2.55\,\text{m s}^{-1}.$$

Check with constant-acceleration formula: $v^2 = 2as = 2 \times 3.24 \times 1 = 6.48$ ✓. Mark scheme: M1 for the work–energy equation, A1 for each correct work term, A1 for the KE expression $\tfrac{1}{2}(16)v^2$, A1 for $v \approx 2.55\,\text{m s}^{-1}$.

(d) Effect of friction in the pulley bearing. (3 marks) If the pulley bearing is rough, the tensions on the two sides of the pulley are no longer equal: the tension on the side of the descending mass B exceeds that on the side of A. Consequently:

- the driving effect on A is reduced, so the acceleration found in (b) decreases;
- part of the work done by gravity is dissipated as heat in the bearing, so the speed found in (c) decreases.

Mark scheme: B1 for stating the tensions differ, B1 for reduced acceleration, B1 for reduced speed / energy dissipated.

SOLUTION **Exercise 12 — A uniform chain sliding off a table** **(14 marks)**

Uniform chain of length $L = 2\,\text{m}$ and mass $M = 4\,\text{kg}$ on a smooth table; initially a length $x_0 = 0.5\,\text{m}$ hangs over the edge and the chain is released from rest. $g = 9.8\,\text{m s}^{-2}$. Linear

density: $\lambda = \dfrac{M}{L} = \dfrac{4}{2} = 2\,\text{kg m}^{-1}$.

(a) Show that $a = \dfrac{gx}{2}$ when a length x hangs over the edge. (4 marks) Mass of the hanging part: λx; its weight $\lambda x g$ is the only external force driving the motion (the table is smooth, and the tension at the bend is internal to the chain). The whole chain moves with the same acceleration a (the chain is inextensible), and the total mass is $\lambda L = M$. Newton's second law for the whole chain:

$$\lambda x g = (\lambda L)\, a \implies a = \frac{x}{L} g = \frac{x}{2} g = \frac{gx}{2}.$$

Note that the acceleration is not constant: it increases as more of the chain hangs over the edge. Mark scheme: M1 for identifying the driving force $\lambda x g$, M1 for applying $F = ma$ to the whole chain of mass M, A1 for the result, A1 for correct use of $\lambda = M/L$.

(b) Speed when the chain just leaves the table ($x = L = 2$ m). (6 marks) Since a depends on position, use $a = v\dfrac{dv}{dx}$:

$$v\frac{dv}{dx} = \frac{gx}{2} \implies \int_0^v v\,dv = \int_{0.5}^{2} \frac{gx}{2}\,dx$$

$$\frac{1}{2}v^2 = \frac{g}{2}\left[\frac{x^2}{2}\right]_{0.5}^{2} = \frac{g}{4}(4 - 0.25) = \frac{g}{4} \times 3.75 = 0.9375g$$

$$v^2 = 1.875g = 1.875 \times 9.8 = 18.375 \implies v = \sqrt{18.375} \approx 4.29\,\text{m s}^{-1}.$$

Mark scheme: M1 for $a = v\frac{dv}{dx}$, M1 for separating variables with correct limits, A1 for the integrated expression, A1 for $v^2 = 18.375$, A1 for $v \approx 4.29\,\text{m s}^{-1}$, B1 for a clear final statement.

(c) Verification by conservation of energy. (4 marks) Take the table top as the reference level for potential energy. The table is smooth, so mechanical energy is conserved. Initial PE: the hanging part has length 0.5 m, mass $\lambda \times 0.5 = 1$ kg, and its centre of mass is 0.25 m below the table: $E_{P,i} = -1 \times g \times 0.25 = -0.25g$. Final PE (chain just vertical, length 2 m, mass 4 kg, centre of mass 1 m below the table): $E_{P,f} = -4 \times g \times 1 = -4g$. Loss of PE: $E_{P,i} - E_{P,f} = -0.25g - (-4g) = 3.75g$. Gain of KE: $\frac{1}{2}Mv^2 = \frac{1}{2} \times 4 \times v^2 = 2v^2$. Conservation of energy: $2v^2 = 3.75g \implies v^2 = 1.875g = 18.375 \implies v \approx 4.29\,\text{m s}^{-1}$, in agreement with part (b). Mark scheme: M1 for the initial PE (centre of mass of hanging part), M1 for the final PE, A1 for the loss $3.75g$, A1 for $v \approx 4.29\,\text{m s}^{-1}$.

11 SUMMARY SHEET

KEY POINTS **Summary Sheet — Connected**

- Key definitions:
  - Connected particles: bodies linked by a light inextensible string.
  - Tension T: force transmitted by the string.
  - Modelling: light inextensible string, smooth surfaces, light smooth pulley.
- Key formulas:
  - Same tension T throughout a light string passing over a smooth pulley.
  - Same magnitude of acceleration for particles linked by an inextensible string.
  - Newton's second law applied to each particle: $F = ma$.
- Methods:
  - Draw a separate force diagram for each particle.
  - Choose a positive direction of motion for the whole system.
  - Write $F = ma$ for each particle, then solve the simultaneous equations for a and T.
- Common mistakes:
  - Forgetting that the tension acts on both particles, in opposite directions.
  - Using different positive directions inconsistently for the two particles.
  - Assuming the tension equals the weight of one of the particles.
- GCE tips:
  - Always state your modelling assumptions (light, inextensible, smooth).
  - Check your answer: the heavier side must accelerate downwards.
  - Give the acceleration and tension to a sensible number of significant figures, with units.

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